Response to the Referee

Dear Editor,

This paper considers an anomaly-free, renormalizable model of an additional Abelian gauge boson A' with chiral couplings to muons and taus. In this model, vector-like leptons and a scalar field S are introduced to construct the ultraviolet completion of the axial $U_A(1)_{L_{\mu-\tau}}$ gauge theory. The phenomenology of the A' in the axial limit is studied, and the relevant experimental constraints are discussed. The paper also discusses a possible future muon-collider probe as a complementary test of the chiral interaction structure.

The referee summarizes our work as follows:

> *The authors construct an anomaly-free, renormalizable model in which an additional Abelian gauge field A' has chiral couplings to muons and taus. This model realizes a limit where the A' coupling to the $\mu - \tau$ current is axial. The second goal of the paper is to examine the phenomenology of this model in the axial limit.*
>
> *The paper presents results that are useful for the model building community. However, I have some significant reservations about key points that either should be included or otherwise justified in being excluded from the manuscript. These are explained below. Revisions for these major points (even if only to clarify why they are not issues) are required before the paper is at the level of a PRD manuscript.*

We sincerely thank the referee for the positive assessment of the model-building construction and for identifying issues that require further clarification or quantitative analysis. We respond to each point below.

Major Points 1).

> *Phenomenology of the S scalar. The second half of the paper reviews a set of experimental bounds on only an effective theory of the A' axially coupled to the $L_{\mu-\tau}$ current. This leaves a major gap between these two parts of the paper: the authors show that their UV theory requires a scalar S that cannot be arbitrarily decoupled*

from the low-energy dynamics, e.g. from Eq. (34) which states that $m_S^2 < 4\pi \left(m_{A'}/(2g_X)\right)^2$.

Reply. The scalar S is not omitted from the UV phenomenology. The manuscript constructs the symmetry-breaking sector, derives the radial-scalar mass and its lepton interactions, and examines the scalar-sector consistency and experimental constraints before specializing the second part to the axial-current observables of A'. Figure 2 is therefore an analysis of the A' effective current on a specified scalar-suppressed benchmark, not a scalar-free replacement of the UV theory.

With the scalar-potential convention

$$V(S) = -\mu_S^2 |S|^2 + \lambda_S |S|^4,$$

the one-VEV gauge and charged-lepton reference construction gives

$$m_{A'} = 2g_X v_s, \qquad m_s = \sqrt{2\lambda_S}\, v_s, \qquad y_{s\ell\ell} \simeq \frac{m_\ell}{v_s} = \frac{2g_X m_\ell}{m_{A'}}, \quad \ell = \mu, \tau.$$

These relations are the UV matching relations of the original one-VEV S+VLL model considered in the manuscript. They are used throughout the scalar and two-parameter benchmark discussions below. No additional scalar field or modified symmetry-breaking sector is assumed in these answers. Thus, at fixed $(m_{A'}, g_X)$, the scalar–lepton couplings are fixed by the symmetry-breaking scale, whereas the physical scalar mass is controlled by λ_S. Equivalently,

$$\lambda_S = \frac{2g_X^2 m_s^2}{m_{A'}^2}, \qquad m_s \leq \sqrt{2\lambda_S^{\max}}\, v_s = \sqrt{\frac{\lambda_S^{\max}}{2}}\, \frac{m_{A'}}{g_X}$$

when the perturbativity criterion $\lambda_S \leq \lambda_S^{\max}$ is imposed.

For the reference choice $\lambda_S^{\max} = 4\pi$ and $m_s \gtrsim 10\,\mathrm{GeV}$, the scalar-suppressed benchmark therefore has the point-dependent range

$$10\,\mathrm{GeV} \lesssim m_s \leq \sqrt{8\pi}\, v_s, \qquad v_s \gtrsim 1.995\,\mathrm{GeV}, \qquad m_{A'} \gtrsim (3.99\,\mathrm{GeV})\, g_X.$$

This is the theoretical consistency condition checked at each point in Fig. 2. The scalar mass is consequently not an unconstrained parameter: its upper value follows from perturbativity once $(m_{A'}, g_X)$ and the scalar benchmark are specified. The mass-dependent scalar searches are discussed separately below.

For clarity, we summarize selected, non-exhaustive low-mass constraints in the scalar-coupling parameterization relevant to this model. The table

separates source-level constraints on a scalar–muon Yukawa interaction from fifth-force limits that act through loop-induced couplings to ordinary matter. The direct scalar entries constrain $y_{s\mu\mu}$ in the normalization specified by their source; they do not constrain g_X independently of the UV matching relation.

For the ultralight binary-pulsar entry, we use Liu and Tang's explicit scalar-mediator analysis, which defines

$$\mathcal{L}_{\rm int} \supset g_{\phi\mu}\phi\bar{\mu}\mu$$

and reports the 95% CL periastron-advance sensitivity in Fig. 5 [1]. With the Yukawa normalization adopted here, $g_{\phi\mu}$ is the same scalar–muon coupling as $y_{s\mu\mu}$, up to the irrelevant overall sign convention. The reported sensitivity covers $10^{-17}\,\mathrm{eV} \lesssim m_\phi \lesssim 2\times10^{-15}\,\mathrm{eV}$, with its best sensitivity, $g_{\phi\mu} < \mathcal{O}(10^{-21})$, at $m_\phi \simeq 4\times10^{-16}\,\mathrm{eV}$. We therefore quote both the mass interval and the best-sensitivity point; the latter is not a uniform upper bound over the entire interval. The $L_\mu - L_\tau$ vector-mediator result of Ref. [2] is not transferred as a scalar-Yukawa limit.

Scalar-mass range	Published constraint	Approximate bound / applicability for $y_{s\mu\mu}$
$10^{-17}\,\mathrm{eV} \lesssim m_s \lesssim 2\times10^{-15}\,\mathrm{eV}$	Binary-pulsar periastron advance (95% CL), ultralight scalar [1]	$\mathcal{O}(10^{-21})$ at the best-sensitivity point $m_s \simeq 4\times10^{-16}\,\mathrm{eV}$; not a uniform bound over the interval
$m_s \lesssim 30\,\mathrm{meV}$	Fifth-force tests through loop-induced scalar–proton coupling [3, 4, 5, 6]	Mass-dependent source constraint; ~ 30 meV is the approximate crossover below which it overtakes astrophysical limits; no universal direct $y_{s\mu\mu}$ limit
$m_s \lesssim 100\,\mathrm{keV}$	Neutron-star cooling [3]	$\mathcal{O}(10^{-13})$; Table 1 gives 7.4×10^{-14} for $m_s \lesssim 1\,\mathrm{keV}$
$m_s \lesssim 1\,\mathrm{keV}$	SN 1987A cooling, free-streaming conservative cold-SN model [3]	$\lesssim 4.2\times10^{-9}$
$100\,\mathrm{keV} \lesssim m_s \lesssim \mathcal{O}(10)\,\mathrm{MeV}$	Diffuse supernova γ-ray background, free-streaming muonic-scalar benchmark [7]	$\lesssim 4.0\times10^{-11}$

These entries describe different observables and source assumptions; they are not combined into a single exclusion band. In Fiorillo et al.'s v2 neutron-star analysis, the scalar sensitivity is mass dependent and extends to approximately 100 keV, with a characteristic reach of $\mathcal{O}(10^{-13})$. The more specific value 7.4×10^{-14} in Table 1 applies to $m_s \lesssim 1\,\mathrm{keV}$, and should not be extrapolated as a constant bound over the full 100 keV interval [3]. The same table gives 4.2×10^{-9} for the conservative cold-SN benchmark at $m_s \lesssim 1\,\mathrm{keV}$. This is a source-specific free-streaming SN 1987A cooling criterion, not a complete trapping or explosion-energy exclusion. The diffuse-supernova-γ-ray

entry is a separate free-streaming muonphilic-scalar benchmark from Caputo et al.: 4.0×10^{-11} for $100\,\mathrm{keV} \lesssim m_s \lesssim \mathcal{O}(10)\,\mathrm{MeV}$, the mass interval treated in their reference analysis [7]. They restrict their muonic-boson treatment to masses $m_\phi \lesssim 10\,\mathrm{MeV}$ for the production and loop approximations used there; we retain that source range rather than extrapolating the quoted value to arbitrarily high masses. It is a bound from the diffuse gamma-ray energy budget, not the same observable as the SN 1987A cooling criterion, and the two entries are not combined.

The fifth-force entry has a different physical origin from the direct scalar–muon limits. In Fiorillo et al.'s muon-only benchmark, the muon triangle induces a scalar–photon interaction and, in turn, a proton coupling

$$g_{\phi p}^{(\mu)} = \frac{\alpha^2}{4} g_{\phi\mu}.$$

Ordinary-matter fifth-force experiments constrain this loop-induced proton coupling, not a tree-level scalar–muon Yukawa vertex. The mass-dependent source curve is shown in Fig. 1 of Fiorillo et al.; it becomes stronger than the astrophysical limits for $m_\phi \lesssim 30\,\mathrm{meV}$. This 30 meV value is the approximate crossover scale quoted in that analysis, not a sharp upper endpoint of fifth-force experimental sensitivity [3]. Thus the fifth-force row is not assigned a single, mass-independent value of $y_{s\mu\mu}$.

In our UV construction, the radial scalar couples to both charged leptons,

$$y_{s\mu\mu} = \frac{m_\mu}{v_s}, \qquad y_{s\tau\tau} = \frac{m_\tau}{v_s}, \qquad \frac{y_{s\mu\mu}}{m_\mu} = \frac{y_{s\tau\tau}}{m_\tau} = \frac{1}{v_s}.$$

The low-momentum scalar–photon form factor therefore contains both charged-lepton loops,

$$\mathcal{F}_{s\gamma\gamma}(q^2) \propto \sum_{\ell=\mu,\tau} Q_\ell^2 \frac{y_{s\ell\ell}}{m_\ell} F_{1/2}\left(\frac{q^2}{4m_\ell^2}\right).$$

In the fifth-force regime $m_s \lesssim 30\,\mathrm{meV} \ll 2m_\mu$, both loop form factors are in their low-momentum limit. Since $Q_\mu^2 = Q_\tau^2 = 1$ and $y_{s\ell\ell}/m_\ell = 1/v_s$, the two contributions have the same sign and add, giving a factor proportional to $2/v_s$. The opposite $U(1)_{L_\mu - L_\tau}$ charges do not cancel this electromagnetic amplitude, which depends on the squared electric charges. Consequently, Fiorillo et al.'s fifth-force curve is quoted here as a muon-only source benchmark, not as a direct $y_{s\mu\mu}$ limit or a complete UV-specific proton-coupling exclusion. A UV translation of this loop-mediated result requires matching both the muon and tau contributions.

The source-level scalar–muon constraints are not invalidated wholesale by $s \to A'A'$. This channel opens only for $m_s > 2m_{A'}$. Where it is open, it must be included in the scalar total width and in the interpretation of observables sensitive to the scalar lifetime, visible branching fractions, cascade topology, or energy deposition. Its existence does not, by itself, remove the source-level constraints listed above.

For a source result that directly constrains the same effective $y_{s\mu\mu}$ normalization, and for a specified m_s within that source's mass range, the one-VEV relation gives

$$y_{s\mu\mu} = \frac{m_\mu}{v_s} = \frac{2g_X m_\mu}{m_{A'}}, \qquad g_X \lesssim \frac{y^{\rm lim}_{s\mu\mu}(m_s) m_{A'}}{2m_\mu}.$$

This is a conditional conversion, not a model-independent bound on g_X. In particular, the muon-only fifth-force curve cannot be inserted into this relation as though it were a direct scalar–muon limit; the μ- and τ-loop matching must be performed first. This is the same conditional conversion given in Eq. (31) of the manuscript, with the source-specific mass dependence of $y^{\rm lim}_{s\mu\mu}(m_s)$ kept explicit.

The table is a source-anchored summary of selected low-mass constraints on the scalar–muon coupling, not a combined scalar–A' exclusion map. Collider searches are discussed separately below because their UV interpretation depends on the portal-mixing benchmark, production rate, total width and lifetime, visible branching fractions, and selection efficiency, including $s \to A'A'$ when kinematically open.

The intermediate-mass scalar phenomenology is also treated explicitly. BaBar and Belle searched for

$$e^+e^- \to \tau^+\tau^- s, \qquad s \to e^+e^-,\, \mu^+\mu^-,$$

over approximately $0.04\,\text{GeV} \lesssim m_s \lesssim 6.5\text{–}7\,\text{GeV}$ [8, 9]. In the mass-proportional benchmark used in these analyses,

$$\mathcal{L} \supset -\xi \sum_\ell \frac{m_\ell}{v_h} s\bar{\ell}\ell, \qquad \xi = \frac{v_h}{v_s}.$$

The Belle result $\xi^{\rm lim} = 0.483$ at $m_s = 1\,\text{GeV}$ [9] therefore gives

$$v_s > \frac{v_h}{\xi^{\rm lim}} \simeq 509\,\text{GeV}, \qquad y_{s\mu\mu} < 2.1 \times 10^{-4},$$

and, using $m_{A'} = 2g_X v_s$,

$$g_X < 9.8 \times 10^{-4} \left(\frac{m_{A'}}{\mathrm{GeV}}\right)$$

at 90% confidence level for this illustrative $m_s = 1\,\mathrm{GeV}$ point with $m_{A'} > m_s/2$, so that the two-body channel $s \to A'A'$ is closed. This example illustrates the mass-dependent scalar-search translation; it is separate from the $m_s \gtrsim 10\,\mathrm{GeV}$ scalar-suppressed reference benchmark used for Fig. 2. If $m_{A'} < m_s/2$, the $A'A'$ mode is open and the visible-channel limit must be interpreted with the UV branching fraction rather than applied as the same g_X bound. For the μ- and τ-lepton couplings relevant to the associated production and dilepton/ditau channels, the mass-proportional benchmark has the same Yukawa normalization as the radial scalar in our construction:

$$y_{s\ell\ell} = \frac{m_\ell}{v_s} = \frac{2g_X m_\ell}{m_{A'}} = \xi \frac{m_\ell}{v_h}, \qquad \xi = \frac{v_h}{v_s}, \qquad \ell = \mu, \tau.$$

At fixed $(m_{A'}, g_X)$, this Yukawa coupling is independent of m_s; changing m_s changes the scalar self-coupling and the kinematically open decay channels, not the mass-proportional Yukawa normalization. The quoted Belle example at $m_s = 1\,\mathrm{GeV}$ uses the dimuon channel. In the minimal construction with negligible portal mixing, the tree-level radial-scalar Yukawa couplings are those to μ and τ; the generic $s \to e^+e^-$ channel in the experimental benchmark is therefore not used as a direct constraint on this UV benchmark.

For $m_s > 2m_{A'}$, the gauge completion predicts the two-vector partial width

$$\Gamma(s \to A'A') = \frac{m_s^3}{32\pi v_s^2} \left(1 - 4x + 12x^2\right) \sqrt{1 - 4x}, \qquad x = \frac{m_{A'}^2}{m_s^2},$$

while each kinematically open charged-lepton channel has

$$\Gamma(s \to \ell^+\ell^-) = \frac{m_\ell^2 m_s}{8\pi v_s^2} \left(1 - \frac{4m_\ell^2}{m_s^2}\right)^{3/2}, \qquad \ell = \mu, \tau.$$

When both channels are open, their ratio is

$$\frac{\Gamma(s \to A'A')}{\Gamma(s \to \ell^+\ell^-)} = \frac{m_s^2}{4m_\ell^2} \frac{(1 - 4x + 12x^2)\sqrt{1 - 4x}}{(1 - 4m_\ell^2/m_s^2)^{3/2}}.$$

Away from the $A'A'$ threshold this ratio can be large, so the $A'A'$ mode can dominate over an individual charged-lepton partial width. The visible

branching fraction is consequently

$$\begin{aligned}\mathrm{BR}(s \to \ell^+\ell^-) &= \frac{\Gamma(s \to \ell^+\ell^-)}{\Gamma_s^{\mathrm{tot}}},\\ \Gamma_s^{\mathrm{tot}} &\supset \Theta(m_s - 2m_\mu)\Gamma(s \to \mu^+\mu^-)\\ &\quad + \Theta(m_s - 2m_\tau)\Gamma(s \to \tau^+\tau^-)\\ &\quad + \Theta(m_s - 2m_{A'})\Gamma(s \to A'A').\end{aligned}$$

Thus the event yield in a dilepton search depends on the production rate, this model-dependent visible branching fraction, and the corresponding selection efficiency. When $s \to A'A'$ is open, the resulting A' decays can also change the final-state topology. The published B-factory limits therefore constrain their stated mass-proportional visible-decay benchmark. Their direct translation to the present UV parameter plane applies where the UV branching fractions reproduce the assumed visible benchmark; when $s \to A'A'$ is open, the interpretation instead requires the model-dependent visible branching fractions and the corresponding final-state selection.

The scalar contribution to the muon anomalous magnetic moment is included together with the axial-vector contribution,

$$\Delta a_\mu = \Delta a_\mu^{A'} + \Delta a_\mu^{s}, \qquad \Delta a_\mu^{s} = \frac{y_{s\mu\mu}^2}{8\pi^2}\int_0^1 dx\, \frac{x^2(2-x)m_\mu^2}{m_s^2(1-x) + x^2 m_\mu^2}.$$

For the reference regime $m_s \gtrsim 10\,\mathrm{GeV}$, the scalar term is below 0.2% of the pure-axial A' contribution. For lighter s, the positive scalar term can instead compete with, or partially cancel, the negative axial contribution. Accordingly, the $(g-2)_\mu$ contour in Fig. 2 is evaluated for the stated scalar-suppressed benchmark, rather than being applied to arbitrary scalar masses.

These results show how the scalar sector is connected to the second part of the paper: its mass and couplings follow from the UV construction; its mass range is restricted by the scalar self-coupling; low-mass and B-factory searches constrain the corresponding scalar couplings; and its contribution to $(g-2)_\mu$ is included when assessing the effective-current benchmark. The second part therefore presents the A' axial-current constraints on a specified, perturbatively consistent scalar benchmark; it does not remove S from the UV theory or treat its phenomenology as absent. The portal benchmark and the associated h–s mixing check are addressed separately under Major Point 1.3.

Major Points 1.1).

> *It is thus an odd choice to exclude the scalar from the phenomenology of the effective theory: if the S is unavoidable at phenomenologically relevant scales, shouldn't one of the main points of the paper be that the only known UV completion of an axial $L_{\mu-\tau}$ gauge boson has a prediction for a scalar particle in a mass range that can be probed by colliders?*

Reply. The scalar is a physical and potentially testable prediction of the UV completion, and its phenomenology is addressed through the model's scalar mass and lepton couplings, perturbativity, and direct-search constraints. In particular, BaBar and Belle searched for $e^+e^- \to \tau^+\tau^- s$, followed by $s \to e^+e^-$ or $\mu^+\mu^-$, over approximately $0.04\,\mathrm{GeV} \lesssim m_s \lesssim 6.5$–$7\,\mathrm{GeV}$ [8, 9]. The mass-proportional benchmark and its translation to v_s, $y_{s\mu\mu}$, and $(m_{A'}, g_X)$ are given in the overall response above. For the present minimal construction with negligible portal mixing, the direct UV mapping uses the dimuon channel, since the tree-level scalar Yukawa relation is $y_{s\ell\ell} = m_\ell/v_s$ for $\ell = \mu, \tau$; the generic $s \to e^+e^-$ search channel is not counted as a direct constraint on this benchmark.

Belle II also provides scalar-specific searches, so its published resonance results in this mass range are not uniformly spin-one-only. The analysis of $e^+e^- \to \mu^+\mu^- X$, $X \to \tau^+\tau^-$, covers 3.6–10 GeV and explicitly tests a Higgs-like spin-zero leptophilic scalar with charged-lepton couplings proportional to their masses [10]. For the μ- and τ-lepton vertices relevant to this production and decay channel, this matches the radial-scalar normalization $y_{s\ell\ell} = m_\ell/v_s = \xi m_\ell/v_h$, with $\ell = \mu, \tau$. The Belle II four-muon analysis covers 0.212–9 GeV and includes a muonphilic-scalar hypothesis in addition to the $L_\mu - L_\tau$ vector boson [11]; that muonphilic benchmark is complementary, but is not identical to the full mass-proportional scalar coupling pattern of our UV construction.

The 3.6–10 GeV tau-pair analysis therefore already supplies a scalar-specific signal hypothesis and selection for the mass-proportional μ- and τ-couplings relevant to the present model. For the radial scalar in our UV completion, however, the additional $s \to A'A'$ channel changes the total width and the visible $\tau^+\tau^-$ branching fraction when it is open. The published scalar coupling or $\sigma \times \mathrm{BR}$ limits must therefore be translated using the UV branching fractions; any $A'A'$-cascade topology also requires its corresponding efficiency. These Belle II results provide relevant scalar sensitivity, but their published

benchmark limits are not, without this translation, a model-independent exclusion of the radial scalar. For context, generic singlet–Higgs mixing and additional-scalar searches have been studied in Refs. [12, 13, 14]. These studies provide complementary information on the scalar sector; the connection to the present UV construction is determined by the model-specific mass, coupling, and decay relations summarized above.

The division of emphasis in the paper is consequently physical: the scalar sector is treated through its UV mass and coupling relations and the scalar-specific constraints, while Fig. 2 is reserved for the axial-current phenomenology of A' in the stated scalar-suppressed benchmark. This keeps the two analyses distinct without omitting the scalar prediction.

Major Points 1.2).

> *What is the range of phenomenologically allowed S mass including perturbativity limits on λ and a motivated range of $m_{A'}$ and g_X. For example: perhaps the authors imagine a range of $m_{A'}$ below some benchmark mass where certain experimental constraints on g_X become very strong. Then with that set of constraints (which should be stated clearly, see minor point #1 below), one may give a range of m_S and comment on the experimental constraints.*

Reply. The manuscript specifies the scalar-mass range point by point through the UV relations, rather than assigning a single scalar mass interval independently of $(m_{A'}, g_X)$. In the one-VEV construction,

$$v_s = \frac{m_{A'}}{2g_X}, \qquad \lambda_S = \frac{m_s^2}{2v_s^2} = \frac{2g_X^2 m_s^2}{m_{A'}^2}.$$

For the scalar-suppressed reference benchmark $m_s \gtrsim 10\,\mathrm{GeV}$ and the perturbativity criterion $\lambda_S \leq 4\pi$, the corresponding point-dependent range is

$$10\,\mathrm{GeV} \lesssim m_s \leq \sqrt{8\pi}\, v_s = \sqrt{2\pi}\, \frac{m_{A'}}{g_X}.$$

It requires

$$v_s \gtrsim \frac{10\,\mathrm{GeV}}{\sqrt{8\pi}} \simeq 1.995\,\mathrm{GeV}, \qquad m_{A'} \gtrsim (3.99\,\mathrm{GeV})\, g_X.$$

This is the theoretical consistency boundary represented by the $v_s \simeq 2\,\mathrm{GeV}$ line in Fig. 2. At each plotted point, v_s is derived and the scalar self-coupling is checked against the perturbativity criterion.

The lower-mass scalar regimes are treated separately through the source-based constraints summarized under Major Point 1 and the direct BaBar/Belle searches discussed under Major Point 1.1. Thus, the manuscript distinguishes the point-dependent perturbative mass range used for the A' benchmark from the mass-dependent scalar searches. The dashed line is a theoretical consistency condition for the specified $m_s \gtrsim 10\,\mathrm{GeV}$ benchmark, not an experimental exclusion or a universal boundary for every possible scalar-sector choice.

Major Points 1.3).

> *Above (34): the authors "assume negligible Higgs-portal mixing." Is this interaction not generated at loop level? It seems that S-h mixing cannot be avoided due to μ and τ loops. In turn, this seems like it would lead to strong experimental bounds on the UV completion. In fact, once there is mixing, is there not a resonance where the mixing becomes large as $m_S \to m_h$? It would be useful to quantify the size of this mixing over the allowed range of m_S. Existing Higgs measurements may set the strongest bounds on the UV completion. If these bounds are obviously negligible (though again, I expect this to be strongly constrained as $m_S \to m_h$), then the authors should present this as the justification for ignoring m_S in the low-energy theory. (Though I am still skeptical that the S can be ignored since it cannot formally decouple.)*

Reply. We agree that a vanishing Higgs-portal coupling is not enforced by the gauge symmetry: the operator $(H^\dagger H)(S^\dagger S)$ is allowed and contributes $(M^2_{\mathrm{loc}})_{hs} = \lambda_{HS} v_h v_s$ after symmetry breaking, so the portal boundary condition must be stated at a matching scale and in a specified renormalization convention. The negligible-portal assumption is therefore used only as a conditional phenomenological benchmark, not as a symmetry-protected statement or a claim that the portal vanishes at every scale.

Generic singlet–Higgs mixing is constrained by the 125 GeV Higgs signal strengths and direct searches for additional scalar states; representative analyses are cited above. For the original one-singlet S + VLL UV model considered here, we complement these general results with the model-specific VLL threshold matching and the finite-momentum h-s self-energy, evaluated using the full charged-lepton/VLL mass-eigenstate matrices as detailed below.

In the $\overline{\mathrm{MS}}$ scheme, with $\mathcal{L} \supset -V$ and $V \supset \lambda_{HS}(H^\dagger H)(S^\dagger S)$, the zero-

momentum effective-potential matching for the vector-like-lepton (VLL) sector gives

$$\lambda_{HS}^{\rm EFT}(\mu_R) = \lambda_{HS}^{\rm UV}(\mu_R) - \sum_{\ell=\mu,\tau} \frac{|y_\ell|^2|\lambda_\ell|^2}{8\pi^2} + \cdots ,$$

where y_ℓ and λ_ℓ are the Yukawa couplings in the VLL interaction. This is a conditional zero-momentum threshold relation; it does not determine the finite-momentum off-diagonal self-energy or a physical scalar mixing angle.

To make the portal assumption explicit, we have evaluated the leading VLL threshold at the matching scale $\mu_M = M_E$,

$$\Delta\lambda_{HS}^{E} = -\frac{1}{8\pi^2}\sum_{\ell=\mu,\tau}|y_\ell|^2|\lambda_\ell|^2, \qquad \lambda_{HS}^{\rm EFT}(\mu_M) = \lambda_{HS}^{\rm UV}(\mu_M) + \Delta\lambda_{HS}^{E}.$$

We use $\lambda_{HS}^{\rm EFT}(M_E) = 0$ as an explicit small-portal reference condition, which requires a nonzero UV portal $\lambda_{HS}^{\rm UV} = -\Delta\lambda_{HS}^{E}$. This condition is a benchmark choice, not a symmetry prediction. In the leading charged-lepton/VLL mass-mixing approximation,

$$\theta_{\ell L} \simeq \frac{y_\ell v_h}{\sqrt{2}M_E}, \qquad \theta_{\ell R} \simeq \frac{\lambda_\ell v_s}{\sqrt{2}M_E}, \qquad m_\ell \simeq \theta_{\ell L}\theta_{\ell R}M_E \simeq \frac{y_\ell\lambda_\ell v_h v_s}{2M_E}.$$

For illustration, taking $M_E = 20\,\text{TeV}$, $\theta_{\mu L} = \theta_{\tau L} = 10^{-2}$, and the leading charged-lepton/VLL mass-mixing relations gives $\Delta\lambda_{HS}^{E} \simeq -0.2108$ at $(m_{A'}, g_X, v_s) = (10\,\text{GeV}, 0.0705, 70.9\,\text{GeV})$, and $\Delta\lambda_{HS}^{E} \simeq -0.01067$ at $(63\,\text{GeV}, 0.1, 315\,\text{GeV})$. For these illustrative points only, we additionally impose $|\lambda_\ell| \le \sqrt{4\pi}$; this is a theory-consistency benchmark rather than a new exclusion contour in Fig. 2. The comparison demonstrates that setting the UV portal to zero is not equivalent to setting the matched EFT portal to zero, and that the latter must be stated as an input if the small-portal benchmark is retained.

The condition $\lambda_{HS}^{\rm EFT}(M_E) = 0$ removes the local dimension-four mixed mass insertion from the matched EFT at the matching scale; it does not set the UV portal to zero. In particular, $\lambda_{HS}^{\rm UV}(M_E) = -\Delta\lambda_{HS}^{E}$ is nonzero in both benchmarks. The CP-even radial-scalar bubbles generated by this portal can be organized using

$$g_{hhh} = 6\lambda_H v_h, \qquad g_{sss} = 6\lambda_S v_s, \qquad g_{hhs} = \lambda_{HS}^{\rm UV} v_s, \qquad g_{hss} = \lambda_{HS}^{\rm UV} v_h.$$

Their zero-momentum-subtracted contribution has the form

$$\Delta\Pi_{hs}^{(h,s)\,\text{subset}} = \frac{\lambda_{HS}^{\rm UV} v_h v_s}{32\pi^2}\left[6\lambda_H\Delta\bar{B}_{hh} + 6\lambda_S\Delta\bar{B}_{ss} + 2\lambda_{HS}^{\rm UV}\Delta\bar{B}_{hs}\right], \qquad \Delta\bar{B}_{ij} \equiv \Delta\bar{B}_0(m_h^2; m_i^2, m_j^2),$$

where the finite, subtraction-scale-independent function is

$$\Delta\bar{B}_0(s;a,b) = -\int_0^1 dx\, \ln\left[1 - \frac{x(1-x)s}{xa+(1-x)b}\right], \qquad a = m_i^2, \quad b = m_j^2.$$

For equal internal masses, this integral has the elementary closed form

$$\boxed{\Delta\bar{B}_0(s;a,a) = 2 - 2\sqrt{\frac{4a}{s}-1}\,\arctan\left(\frac{1}{\sqrt{4a/s-1}}\right), \qquad 0 < s < 4a.}$$

For unequal internal masses, define

$$\Lambda(s,a,b) = (s-a-b)^2 - 4ab, \qquad q = \frac{\sqrt{-\Lambda(s,a,b)}}{2s}, \qquad t_0 = \frac{a-b-s}{2s}, \quad t_1 = t_0 + 1,$$

and

$$G(t) = t\ln\left[\frac{s(t^2+q^2)}{\mu_0^2}\right] - 2t + 2q\arctan\left(\frac{t}{q}\right), \qquad I_0(a,b;\mu_0) = \frac{a\ln(a/\mu_0^2) - b\ln(b/\mu_0^2)}{a-b} - 1.$$

The mixed-mass result is then

$$\boxed{\Delta\bar{B}_0(s;a,b) = I_0(a,b;\mu_0) - \big[G(t_1) - G(t_0)\big], \qquad a \neq b, \quad \Lambda(s,a,b) < 0.}$$

The arbitrary reference scale μ_0 cancels from this difference. All benchmark evaluations below lie on this real sub-threshold branch. Thus the hh and ss terms use the equal-mass expression, while the hs term uses the mixed-mass expression. At $m_s = 100\,\mathrm{GeV}$, the function triplet is $(\Delta\bar{B}_{hh}, \Delta\bar{B}_{ss}, \Delta\bar{B}_{hs}) = (0.1862006, 0.3150601, 0.2366921)$; at $m_s = m_h$, all three equal 0.1862006. The closed forms agree with independent 8,000-interval Simpson integration at relative differences below 10^{-14}. Because $\lambda_{HS}^{\mathrm{UV}}$ is fixed at the one-loop VLL-threshold order, the terms with one portal insertion are next order in this combined counting; the term with two portal insertions is one order further. In an R_ξ gauge, the corresponding Goldstone vertices are $g_{hG_HG_H} = 2\lambda_H v_h$, $g_{sG_HG_H} = \lambda_{HS}^{\mathrm{UV}} v_s$, $g_{haa} = \lambda_{HS}^{\mathrm{UV}} v_h$, and $g_{saa} = 2\lambda_S v_s$. A physical pole calculation must combine these with the gauge/ghost contributions and a consistent tadpole/VEV prescription. Tadpole and VEV-counterterm terms are local and cancel in the displayed momentum difference, but are required in a complete un-subtracted mass/pole renormalization. Thus this expression fixes the perturbative order and diagram structure; it is not by itself a complete two-loop UV–EFT matched self-energy. We do not add this isolated subset to Π_{hs}^{ret} or use it to classify the near-degenerate region. In the gauge basis the singlet radial mode has no direct sWW or $s\bar{t}t$ vertex, so direct one-loop W- or top-quark contributions to Π_{hs} are absent at the retained order; portal-induced effects in those loops enter at higher order.

For each charged-lepton flavour, the field-dependent mass matrix and gauge-basis scalar vertices are

$$\mathcal{M}_\ell(h,s)=\begin{pmatrix}0 & y_\ell(v_h+h)/\sqrt{2}\\ \lambda_\ell(v_s+s)/\sqrt{2} & M_{E_\ell}\end{pmatrix},\qquad \Gamma_h^\ell=\begin{pmatrix}0 & y_\ell/\sqrt{2}\\ 0 & 0\end{pmatrix},\qquad \Gamma_s^\ell=\begin{pmatrix}0 & 0\\ \lambda_\ell/\sqrt{2} & 0\end{pmatrix}.$$

For the τ interaction the original model contains S^*; its radial real-s vertex is the same for the real benchmark couplings. With

$$U_L^T\mathcal{M}_\ell(0,0)U_R=\mathrm{diag}(m_{\ell1},m_{\ell2}),\qquad G_a^\ell=U_L^T\Gamma_a^\ell U_R\quad(a=h,s),$$

all four internal mass-eigenstate pairs are included for both μ and τ. In the following kernels the flavour label ℓ on G_a^ℓ is suppressed.

Define

$$C_{ab}^{ij}=(G_a)_{ij}(G_b)_{ij}^*+(G_a)_{ji}^*(G_b)_{ji},\qquad D_{ab}^{ij}=(G_a)_{ij}(G_b)_{ji}+(G_a)_{ji}^*(G_b)_{ij}^*,$$

and

$$K_{ij}=\bar{A}_0(m_i^2)+\bar{A}_0(m_j^2)+(m_i^2+m_j^2-p^2)\bar{B}_0(p^2;m_i^2,m_j^2).$$

The one-loop fermion self-energy in the source convention is

$$\Pi_{ab}^{F,\ell}(p^2)=-\frac{1}{16\pi^2}\sum_{i,j=1}^{2}\left[C_{ab}^{ij}K_{ij}+2m_im_jD_{ab}^{ij}\bar{B}_0(p^2;m_i^2,m_j^2)\right].$$

We use $\Gamma_{\rm inv}=p^2\mathbf{1}-\mathcal{M}_{\rm loc}^2+\Pi_{\rm RC}$ and $\Pi_B=-\Pi_{\rm RC}$. The finite functions at $\mu_R=M_E$ are

$$\bar{A}_0(m^2)=m^2\left(1-\ln\frac{m^2}{\mu_R^2}\right),\qquad \bar{B}_0(p^2;m_i^2,m_j^2)=-\int_0^1 dx\,\ln\frac{xm_i^2+(1-x)m_j^2-x(1-x)p^2-i0}{\mu_R^2}.$$

The full-theory self-energy contains the local portal-threshold constant and the light-lepton contribution reproduced in the EFT. The momentum-dependent hard term is isolated by

$$\Delta\Pi_{\rm hard}(m_h^2)=[\Pi_B^{\rm full}(m_h^2)-\Pi_B^{\rm full}(0)]-[\Pi_B^{\rm light,exact}(m_h^2)-\Pi_B^{\rm light,exact}(0)],$$

where the light-light term uses the exact light-state Yukawa couplings from the same mass diagonalization. The retained EFT soft contribution is the light-lepton bubble plus the dimension-five contact tadpole,

$$\Pi_{\rm soft,bub+ct}^B(p^2)=\sum_{\ell=\mu,\tau}\frac{q_\ell}{16\pi^2}\left[4\bar{A}_0(m_\ell^2)+(4m_\ell^2-p^2)\bar{B}_0(p^2;m_\ell^2,m_\ell^2)\right],\qquad q_\ell=\frac{2m_\ell^2}{v_hv_s}.$$

We denote their sum by

$$\Pi^{\rm ret}_{hs}(m_h^2) = \Pi^{B}_{\rm soft,bub+ct}(m_h^2) + \Delta\Pi_{\rm hard}(m_h^2).$$

The soft term here is the specified bubble-plus-contact subset, not the complete set of all same-order EFT soft insertions and counterterms. Thus $\Pi^{\rm ret}_{hs}$ is a conditional Yukawa-sector benchmark estimate rather than a fully closed model-level pole prediction.

For $M_E = 20\,\text{TeV}$, $\theta_{\mu L} = \theta_{\tau L} = 10^{-2}$, $m_h = 125.25\,\text{GeV}$, $v_h = 246.22\,\text{GeV}$, and $\mu_R = M_E$, the numerical decomposition is

$[\text{GeV}^2]$	Benchmark A	Benchmark B
$\Pi^{B}_{\rm soft,bub+ct}(m_h^2)$	$-0.437308303 - 0.113173709i$	$-0.098410275 - 0.025468201i$
$\Delta\Pi_{\rm hard}(m_h^2)$	$+0.024030444$	$+0.005407724$
$\Pi^{\rm ret}_{hs}(m_h^2)$	$-0.413277859 - 0.113173709i$	$-0.093002551 - 0.025468201i$

The exact hard momentum difference agrees with the leading source-aligned derivative result to 6.11×10^{-4} relative at both benchmarks, and the zero-momentum hard comparator agrees with the leading portal-threshold estimate to 1.79×10^{-4} relative. These checks validate the stated fermionic subcalculation and its numerical implementation, but do not close the remaining same-order EFT soft/counterterm sectors.

For a clearly non-degenerate point in Benchmark B, we take $m_s = 100\,\text{GeV}$. Here m_s denotes the unmixed diagonal singlet-mass input in the inverse propagator, not an already diagonalized physical pole mass, and

$$\lambda_S = \frac{m_s^2}{2v_s^2} = 0.05039.$$

The singlet width in the mass interval considered is

$$\Gamma_s = \Theta(m_s - 2m_{A'})\frac{m_s^3}{32\pi v_s^2}(1 - 4x + 12x^2)\sqrt{1-4x} + \sum_{\ell=\mu,\tau}\Theta(m_s - 2m_\ell)\frac{m_\ell^2 m_s}{8\pi v_s^2}\left(1 - \frac{4m_\ell^2}{m_s^2}\right)^{3/2}, \qquad x = \frac{m_{A'}^2}{m_s^2}.$$

At this point $s \to A'A'$ is closed because $m_s < 2m_{A'} = 126\,\text{GeV}$, $h \to A'A'$ is closed because $m_h < 2m_{A'}$, and $h \to ss$ is closed because $m_s > m_h/2$. The width is $\Gamma_s = 0.126811\,\text{MeV}$. The first-pass complex-pole solution gives a singlet-like pole at $99.999999997\,\text{GeV}$ with width $0.126815\,\text{MeV}$, and a Higgs-like pole at $125.250000022\,\text{GeV}$ with width $4.069993\,\text{MeV}$.

For this non-degenerate point, the leading-fermionic retained off-diagonal estimate gives

$$\eta_{hs} \equiv \left| \frac{\Pi_{hs}^{\mathrm{ret}}(m_h^2)}{a-b} \right| \simeq 1.695 \times 10^{-5}, \qquad a = m_h^2 - i m_h \Gamma_h, \quad b = m_s^2 - i m_s \Gamma_s.$$

In the isolated-pole, weak-mixing and SM-dominated-channel limit, the mixing-only leading rate diagnostic is

$$\delta\mu_h^{(\mathrm{mix,LO})} \equiv \mu_h^{(\mathrm{mix,LO})} - 1 \simeq -\eta_{hs}^2 \simeq -2.87 \times 10^{-10}.$$

The corresponding mixing-only fractional rate shift is about $2.9 \times 10^{-8}\%$. Here η_{hs} is a first-order off-diagonal mass-insertion diagnostic, not an independently calculated physical mixing angle. The rate estimate is mixing-only: it is not a complete pole signal strength and does not include process-dependent direct-singlet-Yukawa corrections in the μ and τ channels or the separate searches for the 100 GeV scalar.

For comparison, the PDG 2025 update quotes the inclusive global signal strengths $\mu_{\mathrm{ATLAS}} = 1.023^{+0.056}_{-0.053}$ and $\mu_{\mathrm{CMS}} = 1.014^{+0.055}_{-0.053}$ [15]. Their uncertainties are approximately 5.3%–5.6%, more than 1.8×10^8 times larger than the illustrative 2.9×10^{-10} mixing-only rate shift. This is a comparison of scales for the isolated 125 GeV pole in the stated non-degenerate benchmark, not a model-specific likelihood fit or a bound on the separate S-like state. The quoted experimental combinations assume a single narrow resonance and fixed signal shapes; they are not a two-scalar fit. The present comparison is deliberately restricted to a point with a well-separated S-like pole and with $h \to ss$ closed. For $m_s < m_h/2$, scalar-pair decays such as $h \to ss$, when open, depend on the corresponding trilinear coupling and cannot be inferred from $\cos^2\theta$ alone.

The finite-momentum result does not imply uniformly negligible mixing. The relevant complex inverse propagator is

$$\mathcal{D}_B^{-1}(p^2) = \begin{pmatrix} p^2 - m_h^2 + i m_h \Gamma_h & -\Pi_{hs}^B(p^2) \\ -\Pi_{sh}^B(p^2) & p^2 - m_s^2 + i m_s \Gamma_s \end{pmatrix}, \qquad \det \mathcal{D}_B^{-1}(p^2) = 0.$$

The near-pole solver retains the momentum dependence of the light-lepton soft B_0 functions and analytically continues them to the resonance sheet. Its hard part uses the source-aligned finite-derivative approximation; at $p^2 = m_h^2$, replacing this by the exact mass-eigenstate hard difference changes the hard term by 6.11×10^{-4} relative and the total retained Π_{hs} by about 3.43×10^{-5} relative. The diagonal real self-energies are absorbed into the benchmark

mass inputs and the diagonal widths are held fixed, with $\Gamma_h = 4.07\,\mathrm{MeV}$. The resulting poles are therefore first-pass complex-pole diagnostics, not a complete pole-scheme calculation or an experimental lineshape fit.

At exact diagonal-mass degeneracy $m_s = m_h = 125.25\,\mathrm{GeV}$, the two benchmarks give different pole behavior:

	Benchmark A: $A'A'$ open	Benchmark B: $A'A'$ closed
Γ_s	$3.7468\,\mathrm{GeV}$	$0.15894\,\mathrm{MeV}$
$\lvert\Pi_{hs}/(a-b)\rvert$	9.14×10^{-4}	0.19685
$(m_{\mathrm{pole}},\Gamma_{\mathrm{pole}})_{\mathrm{narrow}}$	$(125.250001\,\mathrm{GeV}, 4.0727\,\mathrm{MeV})$	$(125.250041\,\mathrm{GeV}, 0.2921\,\mathrm{MeV})$
$(m_{\mathrm{pole}},\Gamma_{\mathrm{pole}})_{\mathrm{broad}}$	$(125.264006\,\mathrm{GeV}, 3746.39\,\mathrm{MeV})$	$(125.249959\,\mathrm{GeV}, 3.9368\,\mathrm{MeV})$

For the closed-channel benchmark, a $0.1\,\mathrm{MeV}$-step scan over $m_s \in [125.20, 125.30]\,\mathrm{GeV}$ finds a maximum Higgs-like pole-eigenvector composition diagnostic $|s/h| \simeq 0.2039$ near $m_s = 125.2499\,\mathrm{GeV}$; the sampled interval with $|s/h| \geq 0.1$ is approximately 125.2467–125.2532 GeV. In the open-channel benchmark the maximum over the same local scan is 9.15×10^{-4}, and no point reaches 0.1. The quantity $|s/h|$ is a non-Hermitian pole-eigenvector diagnostic, not a directly observable mixing angle or rate.

Separately, the exact mass-eigenstate fixed-self-energy scan over the local interval $m_s \in [10, 150]\,\mathrm{GeV}$, with $1\,\mathrm{MeV}$ spacing and root-refined threshold crossings, gives no interval satisfying $2|\Pi_{hs}/(a-b)| \geq 0.1$ in Benchmark A. In Benchmark B the corresponding diagnostic interval is

$$m_s \in [125.242553544, 125.257446018]\,\mathrm{GeV},$$

or approximately $|m_s - m_h| \lesssim 7.45\,\mathrm{MeV}$. This is a fixed-self-energy diagnostic, not an allowed or excluded interval. The $|s/h|$ eigenvector criterion and $2|\Pi_{hs}/(a-b)|$ criterion are distinct diagnostics and are not identified with one another. In the narrow near-degenerate region, the single-pole approximation $\mu_h \simeq \cos^2\theta$ is not used; a physical rate requires both pole residues, production and decay amplitudes, interference and the experimental mass response.

The perturbativity condition $\lambda_S = m_s^2/(2v_s^2) \leq 4\pi$ gives

$$m_s \leq m_s^{\max} = \sqrt{8\pi}\,v_s = \begin{cases} 355.4\,\mathrm{GeV}, & \text{Benchmark A}, \\ 1579.2\,\mathrm{GeV}, & \text{Benchmark B}. \end{cases}$$

The numerical interval 10–150 GeV is a local diagnostic window around the 125 GeV Higgs pole, not an exhaustive scan of these full perturbative intervals. For $m_s \gg m_h$, at fixed benchmark inputs the retained off-diagonal numerator

varies slowly with m_s, whereas the diagonal mass separation grows as m_s^2; hence the mixing diagnostic decreases parametrically as $|\Pi_{hs}|/m_s^2$. The local scan is not used to infer a global scalar-sector exclusion or viability statement. The radial mode s remains a physical degree of freedom in the UV theory; the scalar-suppressed benchmark only specifies when the axial-current observables in Fig. 2 can be discussed separately from scalar-mediated observables. It does not establish formal decoupling or the absence of scalar effects in every process.

For $q^2 \ll m_s^2, m_{A'}^2$, integrating out s and comparing local exchange coefficients gives scalar four-lepton operators with coefficient scale

$$C_{\ell\ell'}^{(s)} \sim \frac{y_{s\ell\ell} y_{s\ell'\ell'}}{m_s^2}, \qquad y_{s\ell\ell} \simeq \frac{m_\ell}{v_s}.$$

Using $m_{A'} = 2 g_X v_s$, the corresponding A'-exchange coefficient scale is $C^{(A')} \sim g_X^2/m_{A'}^2 = 1/(4v_s^2)$, so

$$\frac{|C_{\ell\ell'}^{(s)}|}{|C^{(A')}|} \sim \frac{4 m_\ell m_{\ell'}}{m_s^2}.$$

For $m_s \geq 10\,\mathrm{GeV}$, this coefficient-level ratio is at most 4.5×10^{-4} for a $\mu\mu$ operator; at $m_s = 100\,\mathrm{GeV}$ it is 4.5×10^{-6}. For a $\tau\tau$ operator it can instead be about 0.13 at $m_s = 10\,\mathrm{GeV}$. This scaling comparison therefore supports separating scalar exchange from the muon-dominated effective-current benchmark, but it does not justify neglecting scalar effects in every τ-sensitive process and is not a process-specific rate comparison.

The one-singlet scalar-suppressed benchmark specifies the regime in which the A' effective-current observables in Fig. 2 are presented separately from scalar-mediated observables; it does not remove s from the UV theory, establish formal decoupling, or imply that scalar effects vanish in every process.

Major Points 2).

> *Motivating the range of effective theory parameters. To further aid in the comparison between the UV theory and the two-parameter low-energy theory, it would be useful to explicitly state in one place (perhaps in a table or a figure like Fig. 2) the range of the effective model parameters that are feasible. The main phenomenological results are in Fig. 2, but it was not clear why this range of couplings and A' masses were chosen: are values outside this range not allowed? What, if anything, does the perturbativity of the UV theory impose on the effective parameters?*

Reply. Fig. 2 focuses on the intermediate-mass region that is the main target of our laboratory phenomenology. Its window brings the low-energy precision, meson-decay, missing-energy, and visible-resonance searches onto a common plot. Lower masses emphasize astrophysical and precision probes, whereas higher masses are increasingly difficult to access with the particular experiments included in this figure. The displayed ranges are

$$10^{-3}\,\mathrm{GeV} \leq m_{A'} \leq 10^{2}\,\mathrm{GeV}, \qquad 10^{-6} \leq g_X \leq 1.$$

They define the plotting window adopted for this comparison, not universal theoretical or experimental boundaries.

Within this window, PIENU probes $m_{A'} < m_{\pi^+} - m_\mu \simeq 33.9\,\mathrm{MeV}$ [16, 17], and the indicative NA62 kaon-decay recast extends to $m_{A'} < m_{K^+} - m_\mu \simeq 388\,\mathrm{MeV}$ [18]. CCFR and CHARM II provide complementary neutrino-trident constraints [19, 20], while NA64$_\mu$ probes the sub-GeV region [21]. BaBar and Belle II cover visible resonances up to approximately 10 and 9 GeV, respectively, and the CMS analysis covers 5–70 GeV [22, 11, 23]. This complementarity, rather than a universal allowed mass interval, motivates the central region displayed.

Below the displayed mass window, astrophysical environments provide important additional sensitivity to weakly coupled light mediators. Supernova cooling and energy transport are particularly relevant to muon- and neutrino-coupled bosons [24, 25]. Their sensitivity also overlaps the sub-GeV part of Fig. 2, so the lower plot edge is not a boundary at which astrophysical constraints suddenly turn on. As explained under Major Point 4.4, the full vector $L_\mu - L_\tau$ exclusion cannot be copied into the axial model because charged-muon production and absorption differ. These astrophysical probes motivate a separate treatment of the lower-mass region rather than extending the laboratory summary plot over many additional decades. Figure 1 supplies this separate ballistic-cooling benchmark: its calculated exclusion does not overlap the displayed $g_X \geq 10^{-6}$, $m_{A'} \geq 1$ MeV window. This statement is specific to the stated cooling criterion and approximations; it does not exclude additional diffusive or explosion-energy constraints.

The more direct reason not to extend the figure to much smaller masses is that the $(g-2)_\mu$ bound already dominates the laboratory constraints displayed at its low-mass end, and its low-mass continuation is fixed by a simple asymptotic behavior. In the scalar-suppressed benchmark, for

$m_{A'} \ll m_\mu$,

$$\Delta a_\mu^{A'} \simeq -\frac{g_X^2}{4\pi^2}\frac{m_\mu^2}{m_{A'}^2}, \qquad g_X^{\lim} \simeq 1.76 \times 10^{-6}\left(\frac{m_{A'}}{1\,\mathrm{MeV}}\right),$$

where the second expression uses the paper's approximate 2σ requirement $\Delta a_\mu^{A'} \gtrsim -8.8 \times 10^{-10}$, obtained from $\Delta a_\mu = 38(63) \times 10^{-11}$ [26, 27, 28]. The boundary therefore continues towards smaller couplings in proportion to the mediator mass, appearing as a straight line in the log–log plane. Extending Fig. 2 further to the left would chiefly prolong this already dominant precision-bound regime, rather than reveal an additional transition among the laboratory probes compared here. We therefore keep the figure focused on the interval that displays their complementary reach. As a numerical illustration, the continuation crosses the plotted coupling floor $g_X = 10^{-6}$ at approximately 0.57 MeV. This reasoning assumes the heavy-scalar benchmark, without a cancellation of the negative axial contribution, and does not establish a universal ordering relative to astrophysical bounds.

Above the displayed window, the on-shell meson-decay, B-factory, and Z-decay channels used here have exhausted their kinematic reach. For low-energy virtual exchange, the relevant interaction strength decreases as $g_X^2/m_{A'}^2$ at fixed coupling, and the heavy-mediator contribution to $(g-2)_\mu$ also decouples as $m_\mu^2/m_{A'}^2$. Thus these measurements have progressively weaker sensitivity to higher masses at the couplings of interest, which motivates ending this comparison at 100 GeV. Heavier mediators remain possible targets for higher-energy experiments, including the future muon-collider tests discussed in the paper; they are not excluded or intrinsically unobservable because they lie outside Fig. 2.

The coupling endpoints 10^{-6} and 1 are likewise display choices: the lower edge retains the low-mass $(g-2)_\mu$ boundary, while the upper edge shows the larger-coupling exclusions. Neither is a strict physical boundary. A conventional gauge perturbativity check is $|g_X Q_i| \lesssim \sqrt{4\pi}$ for every charged field, evaluated at the scale in question. For the unit-charge leptons this gives $g_X \lesssim \sqrt{4\pi}$; including the charge-two scalar gives the stronger estimate $g_X \lesssim \sqrt{4\pi}/2 \simeq 1.77$. The plotted upper edge lies within both estimates, but neither criterion selects $g_X = 1$ as an endpoint. Perturbativity of the independent UV Yukawa and scalar couplings must also be checked for a chosen completion; it does not follow from the gauge coupling alone.

For clarity, the scan variables and their UV interpretation are summarized here:

Quantity	Range or relation	Interpretation
$m_{A'}$	10^{-3}–$10^{2}\,\mathrm{GeV}$	Main laboratory comparison window; lower masses emphasize astrophysical/precision probes, while the searches shown lose reach at higher masses.
g_X	10^{-6}–1	Display window containing the low-mass precision boundary and the larger-coupling exclusions; not a universal allowed interval.
v_s	$v_s = m_{A'}/(2g_X)$	Derived from each selected $(m_{A'}, g_X)$ point, not an independent scan variable.
λ_S	$\lambda_S = m_s^2/(2v_s^2) = 2g_X^2 m_s^2/m_{A'}^2$	Scalar self-coupling in the stated small-portal benchmark; its perturbativity condition must be checked separately.
m_s benchmark	$m_s \geq 10\,\mathrm{GeV}$ for the scalar-suppressed reference case	Reference scalar-mass benchmark used to evaluate the UV consistency condition.

In the scalar-mass convention used in the manuscript, imposing $\lambda_S \leq \lambda_S^{\max}$ gives

$$m_s \leq \sqrt{2\lambda_S^{\max}}\, v_s.$$

For the illustrative choice $\lambda_S^{\max} = 4\pi$, this becomes $m_s \leq \sqrt{8\pi}\, v_s$. Consequently, the reference benchmark $m_s \geq 10\,\mathrm{GeV}$ requires

$$v_s \geq \frac{10\,\mathrm{GeV}}{\sqrt{8\pi}} \simeq 1.995\,\mathrm{GeV}, \qquad m_{A'} \gtrsim (3.99\,\mathrm{GeV})\, g_X.$$

This is the same $v_s \simeq 2\,\mathrm{GeV}$ boundary shown by the dashed line in Fig. 2: the side with $v_s < 2\,\mathrm{GeV}$ does not satisfy the chosen $m_s \geq 10\,\mathrm{GeV}$, $\lambda_S \leq 4\pi$ benchmark. This is a conditional theoretical consistency requirement, not an experimental exclusion. A point that fails it does not support this scalar benchmark; it is not thereby excluded from every possible realization of the UV model if the scalar mass or scalar-sector assumptions are changed. Within the displayed range $g_X \leq 1$, the above gauge-coupling estimates do not set the plot boundary. The scalar self-coupling condition instead imposes a point-dependent check once m_s is specified.

The scalar bounds discussed under Major Point 1 constrain $y_{s\mu\mu}$ directly. They can be translated into the $(m_{A'}, g_X)$ plane only for a specified scalar-

mass benchmark and under the UV relation $y_{s\mu\mu} = 2g_X m_\mu / m_{A'}$; they are not direct, benchmark-independent limits on the pure-axial A' current.

For clarity, Fig. 2 should be read according to the following distinctions:

Region or condition	Interpretation
Excluded by a contour shown in Fig. 2	Excluded under the assumptions of that particular effective-current constraint.
Not excluded by the contours shown	Survives the constraints displayed in Fig. 2; this alone does not establish every UV consistency condition.
Fails the chosen scalar benchmark	Does not satisfy $m_s \geq 10\,\mathrm{GeV}$ together with $\lambda_S \leq 4\pi$ at that point; other scalar-sector choices are not ruled out by this condition alone.
Outside the plotted window	Not forbidden by the scan endpoints themselves; no exclusion or viability claim follows solely from being outside the displayed axes.

The contours therefore communicate the constraints actually applied, while the axis endpoints specify the range chosen to present the phenomenology. The table uses the one-VEV matching relation of the original paper. Its two plotting variables parameterize the original A' benchmark; the table is not a scan over additional UV field content or an enlarged completion parameter space.

Major Points 3).

> *Are neutrino oscillations possible? The Standard Model is defined with only left-handed neutrinos because of the possibility that neutrinos are their own antiparticle. In such a case neutrino masses are generated through, e.g. through the Weinberg operator.*

Reply. Yes. Realistic neutrino masses and mixing can be accommodated by supplementing the axial gauge and charged-lepton construction with the singlet-scalar type-I seesaw used in gauged $L_\mu - L_\tau$ models [29, 30, 31, 32]. The relevant equivalence is that the lepton doublets and the SM Higgs have the same charges in the axial and conventional vector constructions. The reversed charges of the right-handed charged leptons do not enter the neutrino mass operators. The completion uses three right-handed neutrinos and one additional charge-one scalar, together with the existing charge-two scalar S. The selection rules and an explicit mass-generation construction follow.

Reply to Point 3.1

> *In this model, $U_A(1)_{L_{\mu-\tau}}$ invariance allows Weinberg operators for each generation. However, it appears that both the Yukawa matrices and neutrino masses must be aligned and therefore a non-trivial PMNS matrix does not appear to be possible.*

Reply. The charge-two scalar alone leaves a restriction on mixing, but the standard charge-one breaking field removes it. Set $Q_\ell = 1$, absorbing the overall charge normalization into g_X. The original assignments are

$$Q_X(L_e, L_\mu, L_\tau) = (0, 1, -1), \qquad Q_X(e_R, \mu_R, \tau_R) = (0, -1, 1),$$
$$Q_X(H) = 0, \qquad Q_X(S) = 2.$$

If only S obtains a new-symmetry-breaking VEV, the phase transformation $\alpha = \pi$ leaves this VEV invariant and acts as $(+1, -1, -1)$ on the three lepton families. An exact residual Z_2 therefore separates the electron family from the other two.

Define $\chi_\alpha = L_\alpha \cdot H \equiv \epsilon_{ab} L^a_\alpha H^b$, with the Lorentz-spinor contraction in $\chi_\alpha \chi_\beta$ understood. Since $Y(L) = -1/2$ and $Y(H) = +1/2$, this is the electroweak-invariant Weinberg structure in two-component notation. Its $U(1)_X$ charge is $q_\alpha + q_\beta$. The compensating scalar insertions are

Flavor entry	$q_\alpha + q_\beta$	Compensating field	Operator dimension
$ee, \mu\tau$	0	None	5
$e\mu$	$+1$	$\sigma^\dagger$	6
$e\tau$	-1	σ	6
$\mu\mu$	$+2$	$S^\dagger$	6
$\tau\tau$	-2	S	6

The two entries involving σ are absent in the original field content. No integer number of S or $S^\dagger$ insertions compensates their odd charges. Under this S-only symmetry-breaking assumption, the symmetry-allowed EFT mass matrices have the forms

$$M_\nu^{(S)} = \begin{pmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & c & d \end{pmatrix}, \qquad Y_\ell^{(S)} = \begin{pmatrix} y_e & 0 & 0 \\ 0 & y_{\mu\mu} & y_{\mu\tau} \\ 0 & y_{\tau\mu} & y_{\tau\tau} \end{pmatrix}.$$

The off-diagonal μ–τ charged-lepton Yukawa operators are gauge allowed; setting them to zero is the flavor-diagonal benchmark, not a gauge selection

rule. In either case the left-handed rotations retain the electron block and yield $U_{\rm PMNS} = {\rm diag}(1, U_{23})$, up to phases and mass ordering. This need not be the identity, but it cannot describe the observed electron-flavor admixtures. Allowed EFT coefficients alone also do not specify a UV origin of neutrino masses.

Now introduce a complex SM-singlet scalar σ with $Q_X(\sigma) = 1$ and $\langle\sigma\rangle \neq 0$. Its VEV breaks the residual Z_2, and all six independent Majorana entries become allowed. A gauge-invariant EFT parameterization through dimension six is

$$\mathcal{L}_{\nu,\rm EFT} \supset -\frac{1}{2}\sum_{\alpha,\beta} K_{\alpha\beta}\chi_\alpha\chi_\beta + \text{h.c.},$$

$$K = \begin{pmatrix} c_{ee}/\Lambda & c_{e\mu}\sigma^\dagger/\Lambda^2 & c_{e\tau}\sigma/\Lambda^2 \\ c_{e\mu}\sigma^\dagger/\Lambda^2 & c_{\mu\mu}S^\dagger/\Lambda^2 & c_{\mu\tau}/\Lambda \\ c_{e\tau}\sigma/\Lambda^2 & c_{\mu\tau}/\Lambda & c_{\tau\tau}S/\Lambda^2 \end{pmatrix}.$$

This displays the allowed operators, without assuming independent Wilson coefficients in a particular UV completion or a universal ordering of the heavy and symmetry-breaking thresholds. The seesaw below supplies an explicit renormalizable origin. After the extra $U(1)$ is broken and the heavy neutrinos are integrated out, the low-energy theory contains the usual Weinberg operator and light Majorana neutrinos. No light right-handed neutrino is required; heavy right-handed fields in the UV are fully compatible with Majorana light neutrinos.

Reply to Point 3.2

> *Perhaps a full model would require right-handed neutrinos. These would now be charged under $U_A(1)_{L_{\mu-\tau}}$, which likely means the second and third generation must appear as vectorlike combinations. However, it seems to me that mixing between generations is prohibited by $U_A(1)_{L_{\mu-\tau}}$.*

Reply. Introduce three SM-singlet right-handed neutrinos with

$$Q_X(N_{Re}, N_{R\mu}, N_{R\tau}) = (0, 1, -1).$$

These charges allow $\overline{L}_\alpha \widetilde{H} N_{R\alpha}$, with $\widetilde{H} = i\sigma_2 H^*$. At the renormalizable level with a single neutral electroweak Higgs doublet the Dirac matrix is diagonal. We denote the corresponding neutrino Dirac Yukawa couplings by y^ν_α, to

distinguish them from the charged-lepton Yukawa parameters introduced above. The neutral Majorana bilinears admit bare masses M_{ee} and $M_{\mu\tau}$; the two charged singlets supply the other four independent entries. In four-component right-handed-field notation,

$$\begin{aligned}-\mathcal{L}_\nu = & \sum_{\alpha=e,\mu,\tau} y^\nu_\alpha \overline{L}_\alpha \widetilde{H} N_{R\alpha} \\ & + \frac{1}{2} M_{ee} \overline{N^c_{Re}} N_{Re} + M_{\mu\tau} \overline{N^c_{R\mu}} N_{R\tau} \\ & + h_{e\mu} \sigma^\dagger \overline{N^c_{Re}} N_{R\mu} + h_{e\tau} \sigma \overline{N^c_{Re}} N_{R\tau} \\ & + \frac{1}{2} h_{\mu\mu} S^\dagger \overline{N^c_{R\mu}} N_{R\mu} + \frac{1}{2} h_{\tau\tau} S \overline{N^c_{R\tau}} N_{R\tau} + \text{h.c.}\end{aligned}$$

For example, the total charges of $\sigma^\dagger \overline{N^c_{Re}} N_{R\mu}$ and $S^\dagger \overline{N^c_{R\mu}} N_{R\mu}$ are $-1+0+1=0$ and $-2+1+1=0$. All remaining terms are neutral as well. Generation mixing is therefore generated by spontaneous breaking without explicit gauge-symmetry violation.

Take $\langle H^0 \rangle = v_h/\sqrt{2}$, $\langle \sigma \rangle = v_\sigma/\sqrt{2}$, and $\langle S \rangle = v_s/\sqrt{2}$, absorbing VEV phases into couplings. Then

$$m_D = \frac{v_h}{\sqrt{2}} \operatorname{diag}(y^\nu_e, y^\nu_\mu, y^\nu_\tau),$$

$$M_R = \begin{pmatrix} M_{ee} & h_{e\mu} v_\sigma/\sqrt{2} & h_{e\tau} v_\sigma/\sqrt{2} \\ h_{e\mu} v_\sigma/\sqrt{2} & h_{\mu\mu} v_s/\sqrt{2} & M_{\mu\tau} \\ h_{e\tau} v_\sigma/\sqrt{2} & M_{\mu\tau} & h_{\tau\tau} v_s/\sqrt{2} \end{pmatrix}. \tag{1}$$

In the seesaw regime $\|m_D M_R^{-1}\| \ll 1$,

$$M_\nu \simeq -m_D M_R^{-1} m_D^T. \tag{2}$$

All six independent complex entries of M_R are available. For a nonsingular target M_ν and nonsingular diagonal m_D, reconstruct

$$M_R = -m_D^T M_\nu^{-1} m_D. \tag{3}$$

Each reconstructed entry has a gauge-invariant origin in Eq. (1). Substitution into the seesaw formula returns the target matrix. This is a constructive mass-texture existence argument, rather than parameter counting alone.

Neglecting small light–heavy nonunitarity corrections in the stated seesaw and VLL limits, define the light charged-lepton rotation U_L. Then

$$U_\nu^T M_\nu U_\nu = \widehat{m}, \qquad U_{\text{PMNS}} = U_L^\dagger U_\nu,$$
$$\widehat{m} = \operatorname{diag}(m_1, m_2, m_3).$$

For a chosen $U_{\rm PMNS}$, set $U_\nu = U_L U_{\rm PMNS}$ and $M_\nu = U_\nu^* \widehat{m} U_\nu^\dagger$ in Eq. (3). Nonzero light masses with the observed mass-squared differences and mixing can therefore be realized, even for a nondiagonal charged-lepton rotation. The Majorana parameters are independent of the charged-lepton Yukawa parameters, so no alignment is enforced. Perturbativity, heavy-state masses, and flavor constraints still select the physical parameter region; the construction is not a global viability claim for all UV parameters.

The additional fermions preserve anomaly cancellation. In a left-handed Weyl basis their charges are $(0, -1, +1)$, so

$$\Delta\mathcal{A}_{X^3} \propto 0^3 + (-1)^3 + (+1)^3 = 0, \qquad \Delta\mathcal{A}_{\mathrm{grav}^2 X} \propto 0 - 1 + 1 = 0.$$

They are SM singlets and add no mixed SM-gauge anomalies. The opposite-charge pair is vectorlike under $U(1)_X$ before breaking, but the scalar-assisted Majorana entries are allowed after breaking. Adding N_R without σ would still leave the electron block intact.

The correspondence to existing mechanisms is explicit: identify the charge-$(1,2)$ singlets (Φ_1, Φ_2) of Refs. [31, 32] with (σ, S). Their neutrino interactions then have the form above. See Table I and Eqs. (4), (7), (8) of Ref. [31], and Sec. II, Eqs. (1)–(4) of Ref. [32]; our VEV convention fixes the factors of $\sqrt{2}$. This equivalence concerns the neutrino mass sector, not vector-model charged-lepton phenomenology. The single charge-one scalar construction of Ref. [30] instead has $(M_R)_{\mu\mu} = (M_R)_{\tau\tau} = 0$, and hence two zeros in M_ν^{-1} for diagonal m_D. Our existing S allows both entries, so that special two-zero-minor prediction is not imposed here.

Overall response to Major Point 3.

> *Thus it is not obvious to me how neutrino oscillations can be included in this framework. If there is a simple solution, the authors should include it. If there is a known less-simple solution in the literature, the authors should cite this. Otherwise, at best this will be a confusing point. At worst, it would make the construction inconsistent with neutrino oscillation experiments.*

Reply. The two-singlet seesaw provides an established completion with realistic neutrino masses and mixing. The original S-only theory has a residual selection rule; the $\sigma + 3N_R$ completion removes it. The axial charged-lepton assignment itself is therefore not an obstruction to neutrino oscillations. The distinction between these two field contents is essential.

The additional VEV changes the UV parameter relations. With canonical gauge kinetic terms, before the small kinetic-mixing effects considered separately, the scalar kinetic terms give

$$m_{A'}^2 = g_X^2(4v_s^2 + v_\sigma^2), \qquad r_\sigma \equiv \frac{v_\sigma^2}{4v_s^2}. \tag{4}$$

Thus

$$v_s = \frac{m_{A'}}{2g_X\sqrt{1+r_\sigma}}, \qquad y_{s\ell\ell} \simeq \frac{m_\ell}{v_s} = \frac{2g_X m_\ell}{m_{A'}}\sqrt{1+r_\sigma},$$

where the Yukawa expression assumes an approximately S-radial mass eigenstate and small VLL mixing. These relations describe only the optional completion discussed in this neutrino-specific response. They are not used to redefine the original one-VEV model or the scalar, axial-current, kinetic-mixing, finite-vector, and stellar-cooling analyses elsewhere in the response. A small nonzero v_σ removes the exact selection rule, but the mixing magnitudes must still follow from Eq. (1), not from small r_σ alone.

The scalar potential permits

$$\begin{aligned}\Delta V = & -\mu_\sigma^2|\sigma|^2 + \lambda_\sigma|\sigma|^4 + \lambda_{H\sigma}|H|^2|\sigma|^2 \\ & + \lambda_{S\sigma}|S|^2|\sigma|^2 + (\kappa S^\dagger\sigma^2 + \text{h.c.}),\end{aligned}$$

where κ has mass dimension one. These interactions affect the scalar spectrum and mixing; nonzero κ removes the otherwise accidental relative scalar-phase symmetry. The relation $m_s^2 = 2\lambda_S v_s^2$ is a reference limit requiring negligible additional mass shifts and mixing. Moreover, $\sigma\overline{E}_{\mu L}e_R$ and $\sigma^\dagger\overline{E}_{\tau L}e_R$ are gauge allowed. Their coefficients must be sufficiently suppressed to retain the flavor-diagonal charged-lepton benchmark; this is a flavor assumption, not a gauge selection rule.

The two-parameter axial-current results presented elsewhere in this response continue to refer to the original paper's $S+\text{VLL}$ one-VEV model. The $\sigma + 3N_R$ construction is discussed here only as a possible neutrino-sector completion for Major Point 3; it is not incorporated into the UV model used for the other replies, and the original axial-current scan has not been recomputed in that enlarged field content. Applying the phenomenological results to a theory that includes these additional fields would require a separate analysis, which is not part of the present paper or this response.

4).

Is this the full set of experimental constraints? The second half of the manuscript presents a set of experimental bounds on the two-parameter effective A' model. It is not obvious if this is the full set of constraints on the model. (See also point #1 above: I would have liked to see a section discussing the bounds on the S within this model since it cannot be decoupled.) The following experimental bounds seem like they could be relevant. It would be helpful to either include them or justify why they are subdominant to the experiments presented.

Reply.

The contours in Fig. 2 constrain the specified two-parameter pure-axial effective-current benchmark; they are not presented as an exhaustive global classification of every point in the UV model. We address the additional issues raised here by summarizing the scalar constraints under Major Point 1, estimating the momentum-dependent kinetic mixing, quantifying the leading effects of the finite vector component, and distinguishing the supernova contributions governed by the unchanged neutrino current from those sensitive to the charged-muon current. The conclusions below are stated for the explicit benchmark and mass regimes in which the corresponding comparison applies.

4.1).

Neutrino oscillation observations. See point #3 above. It appears that this model predicts a unit PMNS matrix, which is inconsistent with neutrino oscillations.

Reply.

The relevant restriction is the residual-flavor selection rule. If the neutrino mass operators are generated without any additional source of residual-Z_2 breaking beyond the original charge-two scalar S, the symmetry-allowed light-neutrino mass matrix has the $e \oplus (\mu, \tau)$ block form described under Major Point 3. The charged-lepton mass matrix in the same S-only field content has the corresponding block structure. After diagonalizing the charged leptons, the mixing matrix is therefore $U_{\rm PMNS} = \mathrm{diag}(1, U_{23})$, up to phases and mass ordering. This is a conditional statement about the texture allowed by the S-only symmetry-breaking sector, not a neutrino-mass prediction derived in the original charged-lepton construction. It allows a nontrivial μ–τ rotation, but

forbids electron-flavor admixtures and hence cannot reproduce the observed nonzero solar and reactor mixing angles. A renormalizable neutrino-mass completion removes this selection rule while retaining the original axial gauge and charged-lepton construction. We retain S, introduce a complex SM-singlet scalar σ with $Q_X(\sigma) = +1$, and add three right-handed neutrinos with $Q_X(N_R) = (0, +1, -1)$. Since $Q_X(N_{R\alpha}) = Q_X(L_\alpha)$ and $Q_X(H) = 0$, the diagonal Dirac Yukawa terms are gauge invariant. The nonzero σ VEV breaks the residual Z_2 and generates the previously forbidden e–μ and e–τ entries of M_R; the existing S generates the $\mu\mu$ and $\tau\tau$ entries, while the ee and $\mu\tau$ entries are allowed as bare Majorana masses. The resulting general symmetric matrix is given explicitly in Eq. (1). In particular, the axial charges of the right-handed charged leptons do not obstruct this construction: they do not enter the neutrino Dirac or Majorana mass terms.

With

$$m_D = \frac{v_h}{\sqrt{2}} \operatorname{diag}(y_e^\nu, y_\mu^\nu, y_\tau^\nu), \qquad M_\nu \simeq -m_D M_R^{-1} m_D^T,$$

the six independent entries of M_R provide the required flavor structure. The construction demonstrates the compatibility with a general three-flavor mixing matrix explicitly. For a nonsingular target light-neutrino mass matrix, with a chosen nonzero mass spectrum $\widehat{m}$ and observed $U_{\rm PMNS}$, define

$$U_\nu = U_L U_{\rm PMNS}, \qquad M_\nu = U_\nu^* \widehat{m}\, U_\nu^\dagger,$$

where U_L is the left-handed charged-lepton rotation. The reconstruction therefore uses the actual charged-lepton basis and does not require setting U_L to the identity. The additional σ-induced charged-lepton operators are subject to the flavor assumptions specified under Major Point 3. The corresponding heavy-neutrino matrix is then reconstructed as

$$M_R = -m_D^T M_\nu^{-1} m_D.$$

Every entry of this reconstructed matrix has a gauge-invariant origin among the terms in Eq. (1). This is a constructive seesaw realization of the observed mass splittings and three-flavor PMNS mixing, rather than an inference based only on parameter counting. The same construction and its correspondence to the two-singlet-scalar type-I seesaw literature are detailed under Major Point 3 [31, 32].

The optional neutrino completion is also consistent with the gauge sector and anomaly conditions. Within this Major Point 3/4.1 extension only, canonical gauge kinetic terms give the two-VEV relation

$$m_{A'}^2 = g_X^2 (4v_s^2 + v_\sigma^2).$$

This relation belongs only to the optional neutrino completion and is not used in the original-model kinetic-mixing estimate or in any non-neutrino reply. The one-VEV relation is recovered only in the small-r_σ reference regime specified under Major Point 3. In a left-handed Weyl basis the added singlet-neutrino charges are $(0, -1, +1)$, whose linear and cubic sums vanish; the new fermions are SM singlets, and the scalar σ does not contribute to chiral anomalies. The neutrino-oscillation concern is therefore addressed by an explicit, gauge-invariant and anomaly-consistent neutrino-mass completion: the restricted block form characterizes the S-only limit, whereas the $\sigma + 3N_R$ construction accommodates the observed three-flavor mixing.

4.2).

> *There is no quantitative estimate of the amount of kinetic mixing generated with the "benchmark" choices of $\theta \leq 10^{-2}$. The authors make the insightful point that kinetic mixing goes to zero in the axial limit, but because this limit is tied to the size of the mixing (the mass of the vectorlike leptons), for the above benchmark there is non-zero kinetic mixing. It would be useful to have parametric estimates of the amount of kinetic mixing one expects so that readers may compare to experimental bounds. One could invoke parametric expressions in the literature for the relevant loop diagrams.*

Reply.

Finite light–heavy lepton mixing generates a nonzero photon–A' mixing of order $g_X\theta_R^2$. With equal right-handed mixing angles, degenerate physical heavy-lepton masses, and zero independent matching coefficient, $\theta_R = 10^{-2}$ gives $|\epsilon_{\gamma A'}(0)| = 1.44 \times 10^{-6}|g_X|$. At $m_{A'} = 100\,\mathrm{MeV}$ and $g_X = 0.1$, its on-shell value is 1.49×10^{-7}, well below the $\mathcal{O}(10^{-4})$ sensitivity of the NA64 electron-beam invisible search [33]. The derivation and the conditions for this experimental comparison are given below.

We define $\epsilon_{\gamma A'}$ in the photon convention, with induced interaction $\mathcal{L} \supset e\epsilon_{\gamma A'} A'_\alpha J^\alpha_{\mathrm{EM}}$ after diagonalizing the kinetic terms. It is this coefficient, rather than a hypercharge-mixing coefficient before electroweak symmetry breaking, that enters the dark-photon comparison. In the muon block the chiral couplings are $g^\mu_L = g_X$ and $g^\mu_R = -g_X \cos 2\theta_{\mu R}$. Consequently,

$$g^\mu_V = \frac{g^\mu_L + g^\mu_R}{2} = g_X \sin^2\theta_{\mu R}, \qquad g^\mu_A = \frac{g^\mu_R - g^\mu_L}{2} = -g_X \cos^2\theta_{\mu R}.$$

The tau block has the opposite charge signs. At one loop, the vector–axial two-point function vanishes in vacuum: its parity-odd tensor is proportional to $\varepsilon^{\alpha\beta\rho\sigma} q_\rho q_\sigma = 0$. Thus the parity-even photon kinetic term depends on g_V, not on the unsuppressed axial coupling. The photon vertex remains diagonal and proportional to the identity in each light–heavy block, so a single off-diagonal A' vertex cannot close this two-point fermion loop.

The heavy threshold can be exhibited in the same mass basis. Writing $s_\ell = \sin\theta_{\ell R}$, $c_\ell = \cos\theta_{\ell R}$, the four diagonal vector couplings are

$$(g_V^{\mu}, g_V^{\tau}, g_V^{E_\mu}, g_V^{E_\tau}) = g_X(s_\mu^2, -s_\tau^2, c_\mu^2, -c_\tau^2).$$

For these leptons of electric charge -1, the one-loop photon mixing in the above convention can be written as

$$\epsilon_{\gamma A'}(q^2) = \epsilon_{\rm ind} + \frac{e}{2\pi^2} \sum_{f=\mu,\tau,E_\mu,E_\tau} g_V^f \int_0^1 dx\, x(1-x) \ln \frac{m_f^2 - x(1-x)q^2 - i0}{\mu_{\rm r}^2}.$$

Here m_f are physical mass eigenvalues, $\mu_{\rm r}$ is a renormalization scale, and $\epsilon_{\rm ind}$ is the independent local mixing coefficient. Since $\sum_f g_V^f = 0$, the common ultraviolet term and the explicit $\mu_{\rm r}$ dependence cancel in this sum; that cancellation does not fix $\epsilon_{\rm ind}$. We impose the matching condition

$$\boxed{\epsilon_{\rm ind}(\mu_{\rm match}) = 0}$$

at the heavy-VLL matching scale. This boundary condition applies only to the independent local coefficient; it does not set the full momentum-dependent physical mixing $\epsilon_{\gamma A'}(q^2)$ to zero, because finite light-lepton loop contributions remain. For equal right-handed angles the heavy contribution at zero momentum is

$$\epsilon_{\rm heavy}(0) = \frac{e g_X \cos^2\theta_R}{12\pi^2} \ln \frac{m_{E_\mu}^2}{m_{E_\tau}^2}.$$

It vanishes for degenerate physical heavy states; their full loop functions then cancel at any momentum. Equal coupling weights are also required, and equality of the bare mass parameters alone need not ensure exact degeneracy after lepton mixing. We take this paired-heavy-state reference choice together with the matching condition above. These assumptions leave a finite light-lepton contribution because $m_\mu \neq m_\tau$.

The difference from imposing only the original common bare mass M_E can also be estimated. In the small-mixing expansion,

$$m_{E_\ell}^2 \simeq M_E^2(1 + \theta_{\ell L}^2 + \theta_{\ell R}^2), \qquad \epsilon_{\rm heavy}(0) \simeq \frac{e g_X}{12\pi^2}(\theta_{\mu L}^2 - \theta_{\tau L}^2),$$

where the second expression keeps equal right-handed angles and $\epsilon_{\rm ind} = 0$. With $|\theta_{\ell L}| \leq 10^{-2}$, this residual threshold obeys $|\epsilon_{\rm heavy}(0)| \lesssim 2.56 \times 10^{-7} |g_X|$, to leading order in the mixing angles. Its momentum dependence is negligible at 100 MeV for $M_E > 10$ TeV. Thus the physical-mass-degenerate reference and the common-bare-mass choice can be compared without assuming that their thresholds are identical.

For the light-lepton loop, we use the momentum-transfer-dependent kinetic-mixing function studied by Hapitas, Tuckler, and Zhang [34], replacing their charged-lepton vector coupling by $g_X \sin^2\theta_R$ and analytically continuing their spacelike expression with $Q^2 \to -q^2 - i0$. The timelike threshold behavior is compared with Wang and Zhang [25]. Under the matching assumptions above, the remaining contribution is

$$\epsilon_{\gamma A'}(q^2) = -\frac{e\, g_X \sin^2\theta_R}{2\pi^2} \int_0^1 dx\, x(1-x) \log\left[\frac{m_\tau^2 - x(1-x)q^2 - i0}{m_\mu^2 - x(1-x)q^2 - i0}\right],$$

where $\theta_{\mu R} = \theta_{\tau R} \equiv \theta_R$. The light-lepton ultraviolet terms cancel separately in this limit. At $q^2 = 0$, $\int_0^1 dx\, x(1-x) = 1/6$, giving

$$\epsilon_{\gamma A'}(0) = -\frac{e\, g_X \sin^2\theta_R}{12\pi^2} \ln\frac{m_\tau^2}{m_\mu^2} = -\frac{e\, g_X \sin^2\theta_R}{6\pi^2} \ln\frac{m_\tau}{m_\mu}.$$

Using $e = 0.3028$, $m_\mu = 105.658$ MeV, $m_\tau = 1.77686$ GeV, and $\theta_R = 10^{-2}$, the zero-momentum magnitude is

$$\begin{aligned} |\epsilon_{\gamma A'}(0)| &\simeq 1.44 \times 10^{-6} |g_X| \quad (\theta_R = 10^{-2}), \\ &\simeq 1.44 \times 10^{-7} \left(\frac{|g_X|}{0.1}\right) \left(\frac{\theta_R}{10^{-2}}\right)^2 \quad \text{for small } \theta_R. \end{aligned}$$

At fixed off-diagonal lepton mass entry m_R, $\theta_R \simeq m_R/M_E$, so this contribution scales as $g_X m_R^2/M_E^2$. This explicitly exhibits its decoupling with the vector-like-lepton mass under that fixed-parameter choice; maintaining the observed light-lepton masses can require correlated changes of the Yukawa parameters.

For an on-shell A', the relevant timelike argument is $q^2 = m_{A'}^2$. The loop function becomes complex above the charged-lepton pair thresholds; its magnitude is not constant across the scan. A numerical evaluation of the integral gives the following representative values for $\theta_R = 10^{-2}$:

$m_{A'}$	$\lvert\epsilon_{\gamma A'}(m_{A'}^2)\rvert$ / $\lvert g_X\rvert$	$\lvert\epsilon_{\gamma A'}(m_{A'}^2)\rvert$ / $\lvert\epsilon_{\gamma A'}(0)\rvert$
10 MeV	1.44×10^{-6}	1.00
100 MeV	1.49×10^{-6}	1.04
200 MeV	1.82×10^{-6}	1.26
212 MeV	2.12×10^{-6}	1.47
300 MeV	1.74×10^{-6}	1.21
500 MeV	1.40×10^{-6}	0.97
1 GeV	1.08×10^{-6}	0.75
3 GeV	8.06×10^{-7}	0.56
10 GeV	5.47×10^{-8}	3.79×10^{-2}
100 GeV	4.84×10^{-10}	3.35×10^{-4}

The value near $m_{A'} = 2m_\mu$ agrees with the approximately 1.47 enhancement of the magnitude over its zero-momentum value reported by Wang and Zhang [25]. For the low-mass benchmark, the induced electron-current coupling is of order $e|\epsilon| \simeq 4.4 \times 10^{-7}|g_X|$. The momentum-dependent expression should be used for on-shell production and decay; spacelike virtual exchange requires evaluating the same function at the relevant spacelike q^2.

For the reference normalization $g_X = 0.1$, the zero-momentum value is 1.44×10^{-7}, and the largest tabulated on-shell value is 2.12×10^{-7}. The displayed range in Fig. 2 extends to $g_X = 1$, so the corresponding conservative upper estimates over that rectangle are ten times larger. These estimates do not assert that the largest-coupling points satisfy the other constraints.

The appropriate dark-photon comparison is the NA64 electron-beam missing-energy search [33], rather than its thermal-dark-matter parameter combinations. The experiment constrains the electromagnetic production vertex $e\epsilon_{\gamma A'} A'_\alpha J^\alpha_{\mathrm{EM}}$, which is precisely the vertex induced by the photon–A' kinetic mixing here. Its Fig. 3 gives a 90% confidence-level exclusion boundary in the $(m_{A'}, \epsilon)$ plane from 9.37×10^{11} electrons on target. We compare to that mass-dependent boundary, not to the thermal-relic parameters y, α_D, or m_χ.

In the original model, neglecting neutrino masses, the leading widths below the dimuon threshold are

$$\Gamma_{\nu\bar{\nu}} = \frac{g_X^2 m_{A'}}{12\pi}, \qquad \Gamma_{e^+e^-} = \frac{e^2|\epsilon_{\gamma A'}(m_{A'}^2)|^2 m_{A'}}{12\pi}(1 + 2r_e)\sqrt{1 - 4r_e},$$

where the first expression sums the two charged neutrino flavors and $r_e = m_e^2/m_{A'}^2$; the electron width is zero below threshold. In the original model,

this sum runs over the two massless active-flavor channels ν_μ and ν_τ, with $U(1)_X$ charges $+1$ and -1, respectively; the electron flavor is uncoupled. At $m_{A'} = 100\,\mathrm{MeV}$ and $\theta_R = 10^{-2}$, $\Gamma_{e^+e^-}/\Gamma_{\nu\bar{\nu}} \simeq 2.05 \times 10^{-13}$, so $\mathrm{Br}_{\mathrm{inv}} \simeq 1$.

Above the dimuon threshold, the direct axial muon current opens a visible channel. In the same convention,

$$\Gamma_{\mu^+\mu^-} = \frac{m_{A'}}{12\pi}\sqrt{1-4r_\mu}\left[(g_V^\mu)^2(1+2r_\mu) + (g_A^\mu)^2(1-4r_\mu)\right], \qquad r_\mu = \frac{m_\mu^2}{m_{A'}^2}.$$

At $m_{A'} = 300\,\mathrm{MeV}$, with $g_V^\mu = g_X \sin^2\theta_R$, $g_A^\mu = -g_X\cos^2\theta_R$, and $\theta_R = 10^{-2}$, this gives

$$\frac{\Gamma_{\mu^+\mu^-}}{\Gamma_{\nu\bar{\nu}}} \simeq 0.3576, \qquad \mathrm{Br}_{\mathrm{inv}} \simeq 0.7366.$$

The loop-induced electromagnetic widths, including the hadronic width above the two-pion threshold, are negligible relative to these direct widths at the stated benchmark. We therefore compare the missing-energy signal using

$$\epsilon_{\mathrm{eff}}(m_{A'}) = |\epsilon_{\gamma A'}(m_{A'}^2)|\sqrt{\mathrm{Br}_{\mathrm{inv}}(m_{A'})}, \qquad N_{\mathrm{sig}} \propto \epsilon_{\mathrm{eff}}^2,$$

assuming the published missing-energy acceptance for an invisible spin-one state. This is a leading-yield rescaling, not a detector-level recast of visible dimuon decays.

For a transparent mass-dependent comparison, we read the NA64 90% CL boundary from Fig. 3 of Ref. [33] in log–log coordinates. The figure does not provide a machine-readable table; we therefore quote conservative visual-read-off intervals, with an approximately 0.2–0.3 dex line-thickness/read-off uncertainty, rather than spurious high-precision point values. The lower edge of each interval is used in the ratios below. The model values are evaluated from the full timelike loop function at the same mass:

$m_{A'}$	$\lvert\epsilon\rvert/\lvert g_X\rvert$	$\mathrm{Br}_{\mathrm{inv}}$	$\epsilon^{90\%}_{\mathrm{NA64}}$ (visual range)	ϵ_{eff} at $g_X = 0.1$	R at $g_X = 0.1$
10 MeV	1.44×10^{-6}	1.000	$(0.8\text{–}2.0) \times 10^{-5}$	1.44×10^{-7}	1.80×10^{-2}
100 MeV	1.49×10^{-6}	1.000	$(0.5\text{–}1.2) \times 10^{-4}$	1.49×10^{-7}	2.99×10^{-3}
200 MeV	1.82×10^{-6}	1.000	$(0.7\text{–}1.8) \times 10^{-4}$	1.82×10^{-7}	2.59×10^{-3}
300 MeV	1.74×10^{-6}	0.737	$(1.0\text{–}2.5) \times 10^{-4}$	1.50×10^{-7}	1.50×10^{-3}

Here $R \equiv \epsilon_{\mathrm{eff}}/\epsilon_{\mathrm{NA64,low}}$, where $\epsilon_{\mathrm{NA64,low}}$ is the conservative lower edge of the visually read-off interval. The corresponding signal-yield ratios R^2 are 3.3×10^{-4}, 8.9×10^{-6}, 6.7×10^{-6}, and 2.2×10^{-6}, respectively, at $g_X = 0.1$.

Since $\epsilon \propto g_X$, the amplitude ratios are ten times larger and the yield ratios one hundred times larger at $g_X = 1$; even there the largest conservative ratio is $R \simeq 0.18$ at 10 MeV, corresponding to about 3.3% of the limiting signal yield. The other three mass points remain below 10^{-3} of the limiting yield at $g_X = 1$. These upper-edge values are a conservative comparison over the plotted rectangle, not a claim that those large-coupling points satisfy every other constraint.

The secondary-positron resonance energy is $E_{e^+}^{\rm res} = m_{A'}^2/(2m_e)$: it is approximately 0.098, 9.78, and 39.1 GeV at 10, 100, and 200 MeV, respectively, below the approximately 50 GeV missing-energy selection; at 300 MeV it is approximately 88.1 GeV, so the published NA64 curve includes the resonant secondary-positron contribution as well as bremsstrahlung. The rescaling by $\mathrm{Br}_{\rm inv}$ accounts for the open dimuon width at this point. For $g_X = 0.1$ and 1 the total width remains narrow compared with the mass. No dark-matter particle or thermal-relic assumption is required; the invisible final states here are neutrinos.

The residual common-bare-mass threshold estimated above adds at most $2.56 \times 10^{-7}|g_X|$ to the low-momentum mixing, and the corresponding leading-order conservative total at 100 MeV is $1.75 \times 10^{-6}|g_X|$. The mass-dependent table uses the paired physical-heavy-mass reference; the residual-threshold estimate does not change the conclusion that the induced electron-production channel is below the published NA64 sensitivity in these representative mass points. This comparison is separate from the direct muon-production constraint of NA64$_\mu$ in Fig. 2. Unequal right-handed mixing angles, independent heavy-mass splittings, or a nonzero independent matching coefficient require evaluating the general four-state expression above.

4.3).

> *Another way of reframing the above point: the bounds presented are in the pure axial limit. Since the authors present a UV model that only realizes the axial coupling in a decoupling limit, it would be useful to comment on possible experimental bounds on the small vector-like coupling in a model with non-infinite vectorlike fermion masses. If these bounds are always small compared to the axial bounds, this should be stated and justified. The importance of this point is as follows: often bounds are sensitive to the chiral structure of interactions. Sometimes a vector interaction is more tightly constrained than its axial counterpart. Since the authors*

focus on the phenomenology in the axial limit, one wants to make sure that any small-but-finite vector interaction isn't significant because the constraints on the coupling are orders-of-magnitude larger than the axial coupling.

Reply.

We restrict the UV parameter region underlying the axial-vector phenomenology to the finite-mixing domain

$$|\theta_{\ell L}|,\ |\theta_{\ell R}| \leq 10^{-2}, \qquad \ell = \mu, \tau,$$

within the small-mixing benchmark of the UV construction. This is a finite-mass domain, not the strict $M_E \to \infty$ limit: in the seesaw-like regime,

$$m_\ell \simeq \theta_{\ell L}\theta_{\ell R} M_{E\ell}.$$

For the degenerate VLL benchmark, saturating both angle upper values in the τ sector corresponds to $M_E \simeq 17.8$ TeV; smaller mixing angles correspond to larger, still finite VLL masses. In the mass basis, the diagonal light-lepton couplings are

$$g_V^\ell = g_X \sin^2\theta_{\ell R}, \qquad g_A^\ell = -g_X \cos^2\theta_{\ell R}, \qquad \left|\frac{g_V^\ell}{g_A^\ell}\right| = \tan^2\theta_{\ell R} \leq 1.01 \times 10^{-4}.$$

Thus the vector coupling is suppressed by at least four orders of magnitude relative to the axial coupling throughout the parameter region considered. At the reference value $g_X = 0.1$, this gives $|g_V^\ell| \leq 10^{-5}$; at the plotting upper edge $g_X = 1$, it gives $|g_V^\ell| \leq 10^{-4}$. The neutrino couplings remain $g_{\nu_\mu} = +g_X$ and $g_{\nu_\tau} = -g_X$.

The angle restriction also gives a direct criterion for assessing a vector-specific experimental bound. At a fixed $m_{A'}$, let $g_{V,\mathrm{lim}}$ and $g_{A,\mathrm{lim}}$ denote limits on the corresponding vector and axial charged-lepton couplings, obtained with matched production, decay, and acceptance assumptions. Since both couplings are proportional to the same g_X, the vector limit is weaker than the axial limit when

$$\frac{g_{V,\mathrm{lim}}}{g_{A,\mathrm{lim}}} > \tan^2\theta_{\ell R}.$$

Equivalently, the angle required for the vector-specific bound to remain subdominant is

$$|\theta_{\ell R}| < \arctan\sqrt{\frac{g_{V,\mathrm{lim}}}{g_{A,\mathrm{lim}}}}.$$

For our adopted upper value $|\theta_{\ell R}| = 10^{-2}$, a vector-specific limit would have to constrain the vector coupling to below approximately 10^{-4} of the corresponding axial limit—more than four orders of magnitude tighter at the same mass—to yield a stronger bound on g_X. This is the quantitative criterion defining the small-mixing region used here; it does not assert that vector-sensitive limits vanish for arbitrary mixing angles or for unmatched experimental assumptions.

For neutrino trident production, the ratio of the leading Standard-Model interference terms induced by the vector and axial components is parametrically

$$\left|\frac{C_V \delta C_V}{C_A \delta C_A}\right| \sim \frac{C_V}{C_A} \tan^2 \theta_{\mu R} \lesssim 2.0 \times 10^{-4}, \qquad C_V \simeq 0.96, \quad C_A = \frac{1}{2},$$

at the same momentum transfer, before the distinct finite-m_μ response functions are included. For the muon anomalous magnetic moment,

$$\left|\frac{\Delta a_\mu^V}{\Delta a_\mu^A}\right| = \tan^4 \theta_{\mu R} \frac{F_V(r)}{F_A(r)} \lesssim 2.1 \times 10^{-9},$$

using $F_V/F_A \leq 0.20$ over the mass range considered. These are ratios of contributions to the specified observables, not standalone experimental exclusion limits; they show that the finite vector component is subleading in the trident-interference and $(g-2)_\mu$ observables used in our analysis.

For the BaBar, Belle II, and CMS four-muon final-state-radiation searches used in Fig. 2, the unit-coupling vector and axial production kernels become nearly identical for $m_{A'} \gg m_\mu$, with correspondingly similar acceptances; away from threshold, the dimuon widths and branching fractions also approach the same leading behavior [22, 11, 23]. Thus the vector-only coupling reach in these same channels is of the same order as the axial reach, rather than more than four orders of magnitude stronger. In the mixed current, the separate vector-current-squared production contribution is weighted by

$$\left|\frac{g_V^\mu}{g_A^\mu}\right|^2 = \tan^4 \theta_{\mu R} \leq 1.01 \times 10^{-8},$$

in this away-from-threshold regime. The dimuon branching fraction still requires the threshold-dependent width factors given below.

The loop-induced electromagnetic coupling associated with the same vector admixture is treated separately through the kinetic-mixing estimate in our response to Major Point 4.2.

For visible dimuon decays, the vector and axial partial widths have different threshold behavior:

$$\Gamma_V \propto (g_V^\mu)^2(1+2r_\mu)\sqrt{1-4r_\mu}, \qquad \Gamma_A \propto (g_A^\mu)^2(1-4r_\mu)^{3/2}, \quad r_\mu = \frac{m_\mu^2}{m_{A'}^2}.$$

Their ratio is therefore

$$\frac{\Gamma_V}{\Gamma_A} = \tan^4\theta_{\mu R}\,\frac{1+2r_\mu}{1-4r_\mu}.$$

For any chosen tolerance η on the relative vector contribution, the corresponding angle condition is mass dependent:

$$|\theta_{\mu R}| < \arctan\left[\eta^{1/4}\left(\frac{1-4r_\mu}{1+2r_\mu}\right)^{1/4}\right].$$

Away from threshold, the vector-squared contribution is suppressed by $\tan^4\theta_{\mu R} \leq 1.01\times10^{-8}$. Arbitrarily close to $m_{A'} = 2m_\mu$, however, the axial phase-space factor vanishes more rapidly, so no nonzero mass-independent upper bound on the mixing angle can keep the vector-to-axial width ratio uniformly small arbitrarily close to threshold. We therefore retain the pure-axial interpretation of the benchmark observables only where the corresponding threshold factors do not enhance the vector contribution; threshold-sensitive visible-decay interpretations require the full finite-angle widths.

4.4).

> *Are stellar cooling bounds not relevant? See e.g. 2511.09619 for a recent paper on this. If they are not, then this is certainly worth justifying.*

Reply.

Supernova cooling is relevant to the sub-GeV region. The referee-cited analysis is now published as JHEP 06 (2026) 155 [24]; a recent update for the conventional vector $L_\mu - L_\tau$ boson, using the SFHo-18.8 profile, reports SN1987A sensitivity up to $m_{A'} \sim 300$ MeV and low-energy-supernova sensitivity up to about 600 MeV, with SN1987A coupling sensitivity of order 10^{-10} and low-energy-supernova sensitivity reaching 6.4×10^{-12} [25]. These values provide reference scales for assessing the axial benchmark. Kahn et al. give an earlier discussion of stellar cooling for ultralight axial

forces with $m_{A'} < 2m_e$: their Sec. 2.4 emphasizes that an axial A' can have unsuppressed neutrino couplings and a stellar production mechanism not necessarily dominated by kinetic mixing, and that a dedicated analysis is required [35]. This motivates treating the axial case separately, but does not provide a model-specific SN1987A contour for the present $L_\mu - L_\tau$ completion.

In our model, the tree-level neutrino couplings are the same as in the conventional $L_\mu - L_\tau$ model,

$$g_{\nu_\mu} = +g_X, \qquad g_{\nu_\tau} = -g_X.$$

The flavor-summed neutrino-coalescence matrix element and the invisible partial widths $A' \to \nu_\alpha \bar{\nu}_\alpha$ therefore have the same coupling normalization as in the conventional vector $L_\mu - L_\tau$ reference at fixed $(m_{A'}, g_X)$, for the same neutrino phase-space distributions and supernova profile. This follows directly from the original model's active-flavor gauge current $g_{\nu_\mu} = +g_X$, $g_{\nu_\tau} = -g_X$; no additional neutrino fields or completion assumptions are used in this current-level comparison. In the cited vector-current benchmark, neutrino coalescence dominates the production luminosity for $m_{A'} \gtrsim 10$ MeV. For 10 MeV $\lesssim m_{A'} < 2m_\mu$, its SN1987A cooling result therefore provides a useful reference scale, with unchanged neutrino-coalescence production and predominantly invisible decays. This quoted channel hierarchy is specific to the vector benchmark; the axial calculation below evaluates the competition with semi-Compton production explicitly. The published vector curve is retained as a reference, separately from our stated axial-EFT cooling benchmark.

The distinction arises in processes involving the charged-muon current. The semi-Compton process $\mu\gamma \to \mu A'$ and its inverse absorption depend on whether that current is vector or axial. For on-shell muons, the vector current is conserved whereas the axial current obeys

$$q_\alpha \bar{u}(p')\gamma^\alpha u(p) = 0, \qquad q_\alpha \bar{u}(p')\gamma^\alpha \gamma_5 u(p) = 2m_\mu \bar{u}(p')\gamma_5 u(p), \qquad q = p' - p.$$

The non-conserved axial current therefore contributes to longitudinal A' emission. In the relativistic longitudinal limit $E_{A'} \gg m_{A'}$, Goldstone equivalence gives the effective coupling scale

$$|g_{G\mu}| \sim \frac{2|g_A^\mu| m_\mu}{m_{A'}}.$$

The semi-Compton emissivity and inverse absorption must consequently be evaluated with the axial current and its polarization structure; the vector-model curve is not directly transferable in the low-mass region where semi-Compton production dominates. Above $m_{A'} = 2m_\mu$, the axial and vector

dimuon widths also differ in their threshold dependence, as quantified under Major Point 4.3, affecting visible-decay and low-energy-supernova energy-deposition constraints.

The low-momentum kinetic-mixing estimate has a separate scope. In the equal-mixing benchmark,

$$\frac{|\epsilon_{\gamma A'}(0)|}{|g_X|} \simeq 1.4 \times 10^{-6},$$

compared with approximately 1/70 in the conventional vector model. At the same vacuum kinematics, this suppresses only electromagnetic-current-mediated amplitudes and rates, such as plasmon conversion or the mixing-induced proton coupling in np bremsstrahlung, by about 10^{-4} and 10^{-8}, respectively. It does not suppress neutrino coalescence or the direct axial semi-Compton process. These factors are same-kinematics vacuum scaling estimates, not supernova emissivity ratios; an actual in-medium rate requires the momentum-dependent mixing and plasma propagators.

An explicit axial-EFT cooling benchmark. We evaluate the axial interaction at the effective-current level, with $g_A^\mu = -g_X$, $g_V^\mu = 0$, and the active-neutrino currents above. An identical ultraviolet completion is not required to reuse a calculation with these operators. Existing axial-vector cooling estimates use longitudinal-mode matching [36, 37]; here finite-mass semi-Compton scattering is obtained from the exact tree-level spin-one kernels of Liu and Miller [38], with the electron mass and coupling replaced by those of the muon. The small vector admixture described under Major Point 4.3 can be retained in this EFT; the plotted benchmark is its pure-axial limit.

For $\gamma(k)\mu(p) \to A'(q)\mu(p')$, define $M = m_{A'}$, $m = m_\mu$, $S = (p+k)^2 - m^2$, $U = (p-q)^2 - m^2$, and $t = (p'-p)^2$. Then

$$S + U + t = M^2, \quad R = \frac{(S+U)^2}{SU}, \quad B = m^2 \left(\frac{S+U}{SU}\right)^2 - \frac{t}{SU},$$

and the initial-spin-averaged, final-polarization-summed result is

$$\overline{|\mathcal{M}|^2} = e^2\left[(g_V^\mu)^2 \mathcal{A}_V + (g_A^\mu)^2 \mathcal{A}_A\right], \quad \mathcal{A}_V = 4 - 2R + 4(M^2 + 2m^2)B,$$

$$\mathcal{A}_A = 4 - \left(2 + \frac{4m^2}{M^2}\right) R + 4(M^2 - 4m^2)B, \qquad \frac{d\sigma}{dt} = \frac{\overline{|\mathcal{M}|^2}}{16\pi S^2}.$$

The VA interference vanishes for this unpolarized vacuum $2 \to 2$ observable. These expressions retain the full masses and do not use the high-energy Weizsäcker–Williams approximation. An independent explicit-spinor evaluation agrees with the published kernels in 28 pure/mixed-current comparisons, with maximum relative difference 2.9×10^{-12}.

For transport we keep each mediator polarization $h = L, +, -$ separately. Inverse Compton scattering is evaluated with an initial muon at rest, exact two-body recoil, and the effective muon density

$$n_\mu^{\rm eff}(r) = 2 \sum_{\eta=\pm1} \int \frac{d^3p}{(2\pi)^3} f_\mu(E; \eta\mu_\mu)\,[1 - f_\mu(E; \eta\mu_\mu)].$$

Writing $k_A = \sqrt{\omega^2 - M^2}$, $D = m + \omega - k_A c$, the outgoing photon energy and differential absorption kernel are

$$\omega_\gamma = \frac{M^2 + 2m\omega}{2D}, \qquad v_A \frac{d\sigma_h^{\rm abs}}{d\Omega} = \frac{\overline{|\mathcal{M}_h^{\rm abs}|^2}\,\omega_\gamma}{64\pi^2 m\omega D}.$$

Here only the initial muon spin is averaged. Multiplication by $n_\mu^{\rm eff}$, angular integration and the factor $1 + f_B(\omega_\gamma)$ give $\Gamma_h^{\rm abs}$. Production is assigned through thermal detailed balance,

$$\Gamma_h^{\rm prod} = e^{-\omega/T}\Gamma_h^{\rm abs}, \qquad \Gamma_h^{\rm red} = (1 - e^{-\omega/T})\Gamma_h^{\rm abs}.$$

This is the stated muon-rest/effective-density approximation, with exact recoil; it is not an exact moving-muon thermal integral.

Neutrino production and decay are computed with Fermi–Dirac occupations and Pauli blocking. For each flavor, $\Gamma_{\nu,0} = g_X^2 M/(24\pi)$, and

$$\Gamma_{\nu,h}^{\rm prod/abs} = \frac{M}{\omega}\Gamma_{\nu,0} \int_{-1}^{1} dz\, W_h(z) \begin{cases} f_\nu(E_+; \mu_\nu) f_{\bar\nu}(E_-; -\mu_\nu), \\ [1 - f_\nu(E_+; \mu_\nu)][1 - f_{\bar\nu}(E_-; -\mu_\nu)], \end{cases}$$

$$E_\pm = \frac{\omega \pm k_A z}{2}, \quad W_L = \frac{3}{4}(1 - z^2), \quad W_\pm = \frac{3}{8}(1 \pm z)^2, \quad \int_{-1}^{1} W_h\, dz = 1.$$

The two flavors are summed with their respective chemical potentials, taking $\mu_{\nu_\tau} = 0$. In particular, a nonzero μ_{ν_μ} can distinguish the two circular transverse rates, so their opacities are not averaged before exponentiation. Above $M = 2m_\mu$, dimuon inverse decay and decay are included with the axial width $g_X^2 M(1 - 4m_\mu^2/M^2)^{3/2}/(12\pi)$, the appropriate massive two-body kinematics and muon blocking factors.

We use a reconstructed SFHo-18.8 profile at one second after bounce. The temperature and muon/neutrino chemical potentials are extracted from the published vector-graphics figure in Ref. [7]; the gravitational lapse is extracted from Ref. [39]. These are digitized published profiles, not the original simulation tables. Reconstructing the muon density from the extracted T, μ_μ agrees with the independently plotted density at five radii between 0 and 15 km to approximately 0.5–1.1%. The two published temperature curves agree within 0.27% over their common range. The last kilometre of the lapse is continued by an exterior Schwarzschild form matched at 24 km; unresolved sub-MeV neutrino chemical potentials are set to zero.

Following the ballistic-cooling prescription of Ref. [24], the luminosity at infinity is

$$L_{A'} = \sum_h \int_0^{R_\nu} 4\pi r^2 dr\, N(r)^2 \int_{M/N(r)}^{\infty} d\omega\, \frac{k_A \omega^2}{2\pi^2}\, \Gamma_h^{\rm prod}(r,\omega)\, \mathcal{P}_h(r,\omega),$$

$$\mathcal{P}_h = \frac{1}{2}\int_{-1}^{1} dc_\theta\, \exp\!\left[-\int_0^{s_{\max}} \frac{ds}{v_A}\, \Gamma_h^{\rm red}\!\left(\sqrt{r^2+s^2+2rsc_\theta}, \omega\right)\right],$$

$$s_{\max} = -rc_\theta + \sqrt{R_\nu^2 - r^2 + r^2 c_\theta^2}, \qquad R_\nu = 25 \text{ km}.$$

This includes inward and outward trajectories, the velocity factor, stimulated-emission reduction of opacity, gravitational redshift of luminosity and the escape-energy threshold. As in this prescription, the local energy is held fixed along a straight chord; this is not full general-relativistic ray transport. The polarization sum is outside the escape exponential. The cooling criterion is $L_{A'} > 4.4 \times 10^{52}\,\mathrm{erg\,s^{-1}}$, with the alternative $3.0 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ also shown.

Representative intervals above the $4.4 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ criterion are

M [MeV]	$\lvert g_X\rvert_{\rm lower}$	$\lvert g_X\rvert_{\rm upper}$
1	3.8×10^{-11}	4.9×10^{-7}
10	3.6×10^{-10}	7.5×10^{-8}
100	2.4×10^{-10}	7.3×10^{-9}
200	4.1×10^{-10}	2.9×10^{-9}

and the benchmark closes near $M \simeq 0.26$ GeV. The lower edge is production limited; the upper edge includes absorption and decay trapping. At $M = 10$ MeV, semi-Compton emission supplies approximately 91% of the optically thin axial luminosity, whereas neutrino coalescence supplies approximately 99.4% at 100 MeV. Thus the vector benchmark's quoted dominance above approximately 10 MeV cannot be assumed unchanged for the axial current.

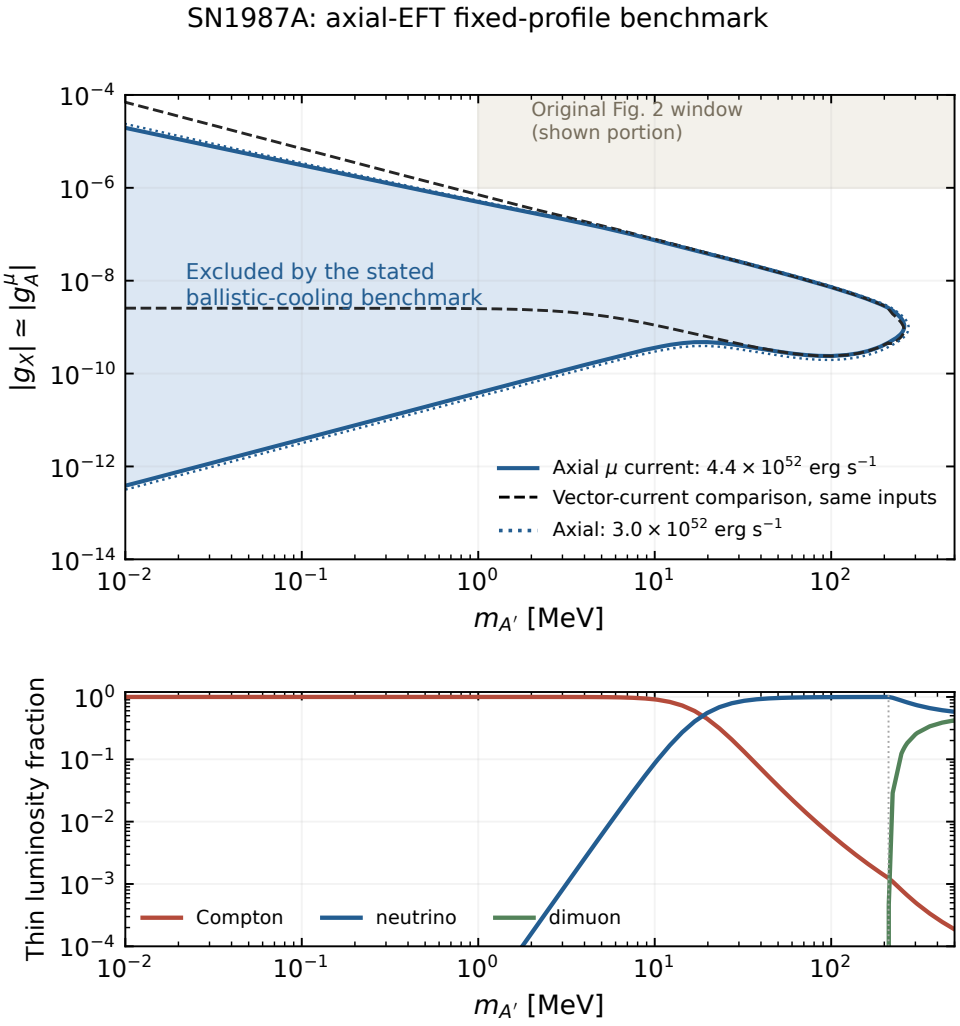


Figure 1: Conditional SN1987A ballistic-cooling exclusion for the axial EFT. Top: the blue region exceeds the stated $4.4 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ criterion; dotted blue uses $3.0 \times 10^{52}\,\mathrm{erg\,s^{-1}}$. The dashed curve is a vector-current comparison computed with the same profile, channels and approximations, not a copied published contour. The beige area shows the displayed portion of the original Fig. 2 window. Bottom: contributions to the optically thin axial luminosity. The calculation includes semi-Compton scattering, neutrino coalescence/decay, and dimuon coalescence/decay. It assumes vacuum photon dispersion, a reconstructed stationary profile and the muon-rest/effective-density treatment; unshaded regions are not established globally allowed regions.

Several checks determine the precision of this comparison. The kernels reproduce the vector Thomson limit, the axial longitudinal mass scaling, the vacuum neutrino/dimuon widths and mode-specific detailed balance. The angular escape calculation reproduces the analytic uniform-sphere result. Increasing the radial, energy and angular quadrature gives stable lower edges at the sub-percent level; the most sensitive low-mass upper edge changes by approximately 8% in the refinement checks, while the 1–250 MeV upper edges are substantially more stable. These numerical checks are separate from physical approximation uncertainties. Scaling the semi-Compton rate and its inverse together by factors 1/2 and 2 moves the low-mass lower edge by approximately +41% and –29%, respectively. This is a sensitivity test, not a confidence interval. Changing the common source/escape radius to 20 km changes the upper edge by approximately a factor of two at 10 keV and by 25% at 100 MeV; setting $\mu_{\nu_\mu} = 0$ changes the tested boundaries by less than 1%.

In the relativistic longitudinal limit the result can also be checked against the corrected ALP–muon bound of Bollig et al. [40, 41]:

$$\widehat{g}_{a\mu} = \frac{|g_A^\mu|}{M}, \qquad \widehat{g}_{a\mu} \lesssim 10^{-7.5}\ \mathrm{GeV}^{-1}.$$

For the $3.0 \times 10^{52}\,\mathrm{erg\,s^{-1}}$ criterion our low-mass lower edge is approximately

$$|g_A^\mu|_{\mathrm{lower}} \simeq 3.2 \times 10^{-11}(M/\mathrm{MeV}),$$

consistent with that longitudinal estimate. The finite-mass calculation and its upper edge do not follow from extending this straight line.

The plotted exclusion is specifically a stationary-profile ballistic-cooling benchmark. Muon thermal motion and full finite-temperature photon dispersion are not included; muon–proton bremsstrahlung and other additional channels are not part of this three-channel estimate. The approximate rate rescaling above tests sensitivity without replacing a calculation of those effects. Sufficiently strong new interactions can also change the underlying supernova profile. The diffusive-transport, high-energy-neutrino and LESN observables discussed in Ref. [24] impose different criteria and are not represented by this shading. Within the stated benchmark the ballistic exclusion lies below the $g_X \geq 10^{-6}$, $M \geq 1$ MeV portion displayed in the original Fig. 2. This explains why that laboratory-summary window and the new cooling figure cover complementary couplings; it does not establish that astrophysical energy transport is irrelevant within the original window.

Thus stellar constraints are not dismissed: the neutrino-coalescence contribution is applicable where it dominates, while the full vector-current supernova exclusion curve is not treated as a universal exclusion of the axial model. The semi-Compton and electromagnetic-current-mediated channels are distinguished according to their respective interactions, preserving the relevant astrophysical information without applying a bound outside the interaction structure for which it was derived.

Minor Point 1).

> *Range of parameters. The general construction of the model is clearly written, but it is difficult for the reader to envision the mass and coupling range that the authors are targeting. It would be very helpful if the authors decide on a range of $m_{A'}$ and g_X of interest. For example: is there a range of masses where experimental or theoretical limits are less constraining? Is there some set of phenomenology that the authors seek to realize with a "light" A'? If so, they should define "light." (See, e.g. the first word of the introduction, which is not defined with respect to any reference mass scale.)*

Reply. The word "light" is not intended to specify a separate quantitative mass regime. The Abstract describes the broader MeV-to-TeV scale, while Fig. 2 displays the specific comparison window

$$10^{-3}\,\mathrm{GeV} \leq m_{A'} \leq 10^{2}\,\mathrm{GeV}, \qquad 10^{-6} \leq g_X \leq 1.$$

The numerical range and the observables included in the analysis, rather than the adjective "light", define the phenomenological domain considered here. The motivation for this window and the complementary experimental coverage are detailed in our response to Major Point 2.

Minor Point 2).

> *Defining the pure-axial limit. Equation (12) gives the coupling of the A' to the mass-basis fields. It is here that the "pure axial coupling" limit is defined? i.e. $\theta_{\mu R} \to 0$? If that is the case, it would help to define this as the pure-axial limit, a phrase which is used below (36) but not otherwise quantified.*

Reply.

We agree that the pure-axial limit should be quantified when the mass-basis couplings are introduced in Eq. (12). In the convention used there, the strict pure-axial limit is $\theta_{\mu R} \to 0$:

$$g_V^\mu = g_X \sin^2\theta_{\mu R} \to 0, \qquad g_A^\mu = -g_X \cos^2\theta_{\mu R} \to -g_X.$$

For finite $\theta_{\mu R}$, the vector coupling is nonzero, so the interaction is not strictly purely axial. We therefore reserve the term "pure-axial limit" for $\theta_{\mu R} \to 0$; the analogous limit in the τ sector follows with the charge reversal specified in the model.

Minor Point 3).

> *Bounds on the vectorlike leptons. Because the mixing angles scale inversely with the vectorlike lepton masses, it would be helpful to present lower limits on these masses. This, in turn, gives an upper limit on the $\theta_{(\mu,\tau)R}$ mixing angles which then gives an upper limit on the A' vectorlike coupling to $L_{\mu-\tau}$. How do these bounds compare with the $\theta_{(\mu,\tau)(L,R)}$ bounds in equation (27)?*

Reply.

In the decoupling regime, the light–heavy mixing angles scale inversely with the vector-like-lepton masses at fixed off-diagonal mass parameters. After matching to the observed charged-lepton masses, Eqs. (21) and (23) give

$$m_\ell \simeq \theta_{\ell L}\theta_{\ell R} M_{E\ell}, \qquad \ell = \mu, \tau.$$

Combining these relations with the small-mixing condition adopted in Eq. (27),

$$|\theta_{\ell L}|,\ |\theta_{\ell R}| \leq 10^{-2},$$

gives the following conditional mass scales at leading order:

$$M_{E\mu} \gtrsim \frac{m_\mu}{(10^{-2})^2} \simeq 1.06 \text{ TeV}, \qquad M_{E\tau} \gtrsim \frac{m_\tau}{(10^{-2})^2} \simeq 17.8 \text{ TeV}.$$

For the degenerate choice $M_{E\mu} = M_{E\tau} \equiv M_E$ in Eq. (22), the tau relation therefore gives the stronger conditional requirement $M_E \gtrsim 17.8$ TeV. This follows from imposing the mixing-angle condition and reproducing the charged-lepton masses simultaneously; it is not an independent direct-search mass

limit. The separate condition $M_E > 10$ TeV in Eq. (22) is weaker than this combined requirement.

Equation (27) and the mass relations play different roles: Eq. (27) bounds each mixing angle individually in the adopted small-mixing setup, whereas the mass relations fix the products $\theta_{\ell L}\theta_{\ell R}$. A mass relation by itself would not yield an individual upper bound on $\theta_{\ell R}$. Using the Eq. (27) condition in the mass-basis couplings,

$$g_V^\mu = g_X \sin^2\theta_{\mu R}, \qquad g_A^\mu = -g_X \cos^2\theta_{\mu R},$$
$$g_V^\mu \lesssim 10^{-4} g_X,$$
$$\left|\frac{g_V^\mu}{g_A^\mu}\right| = \tan^2\theta_{\mu R} \lesssim 10^{-4}.$$

Thus, the residual vector component is suppressed by at least four orders of magnitude relative to the axial component under the adopted condition. This is a consequence of the small-mixing assumption, not a separate experimental bound; by itself it does not establish that every observable is insensitive to the vector component. We clarify the terminology of Eq. (27) in our response to Minor Point 4.

Minor Point 4).

> *Conservative benchmark. Equation (27) is presented as a "conservative benchmark." What does this mean? A benchmark point would be a choice of these mixing angles. A constraint is an upper bound. Eq. (27) is written as an upper bound, but is described as a benchmark. This is not simply semantic: readers would want to understand if* 10^{-2} *is an experimental upper limit or simply a convenient choice.*

Reply. Equation (27) specifies the small-mixing assumption used in the phenomenological analysis; it is neither a benchmark point nor a universal experimental upper limit on all four mixing angles. Under this assumption, mixing-dependent corrections are suppressed at order $\theta^2 \lesssim 10^{-4}$, and the residual vector component of the A' coupling is correspondingly small. The bounds on vector-like leptons depend on their electroweak representation and on the observables considered, so the value 10^{-2} defines the controlled small-mixing regime rather than a direct experimental limit.

Minor Point 5).

What is the purpose of including muon collider bounds? The second half of the paper focuses on existing experimental bounds on the effective theory. It is a strange change in tone to include hypothetical muon collider bounds for future reach. The authors do not justify why a future muon collider is the way to test this model in the future and the section feels like it is tacked on. Most of the section emphasizes that the muon collider search would be a "standard" follow up of existing work (Ref 83) and so this section does not seem particularly novel. The authors simply focus on the forward-backward asymmetry. Why not also include polarization asymmetries? Or would there be asymmetries in the ee versus $\tau\tau$ final states? Given the first half of the paper, though, it seems that the real "smoking gun" for this UV completion is the physics of the S particle which should be tested in any Higgs-factory phase of a future lepton collider. I feel that this section either adds too little and should be downplayed as an example of future tests, or otherwise should have its own full fledged exploration (which may be a separate paper).

Reply. A high-energy muon collider is a natural complementary probe because A' couples directly to the initial-state muons and can therefore be tested in the heavier-mass regime, beyond the low-energy probes emphasized in the first part of the paper. Ref. [42] provides a useful vector $L_\mu - L_\tau$ collider benchmark at $\sqrt{s} = 3\,\mathrm{TeV}$. The axial model, however, cannot be assessed by simply importing the vector result: the left- and right-handed couplings entering the helicity amplitudes are different.

We use the $\tau^+\tau^-$ final state because it receives only s-channel γ, Z, A' exchange; the same-flavor $\mu^+\mu^-$ final state also contains t-channel diagrams, which complicate the angular interpretation. For $\mu^+\mu^- \to \tau^+\tau^-$, neglecting the external-lepton masses, the unpolarized angular distribution is

$$\frac{d\sigma_{\tau\tau}}{d\cos\theta} = \frac{s}{128\pi}\Big[\left(|\mathcal{A}_{LL}|^2 + |\mathcal{A}_{RR}|^2\right)(1+\cos\theta)^2 + \left(|\mathcal{A}_{LR}|^2 + |\mathcal{A}_{RL}|^2\right)(1-\cos\theta)^2\Big],$$

where the chiral amplitudes are

$$\mathcal{A}_{ij} = \frac{e^2}{s} + \frac{g_i^Z g_j^Z}{s - m_Z^2 + i m_Z \Gamma_Z} + \frac{g_i^\mu g_j^\tau}{s - m_{A'}^2 + i m_{A'}\Gamma_{A'}}, \qquad i,j = L,R,$$

and the Standard Model Z-boson couplings are

$$g_Z = \frac{e}{s_W c_W}, \qquad g_L^Z = g_Z\left(-\frac{1}{2} + s_W^2\right), \qquad g_R^Z = g_Z s_W^2.$$

Here i labels the initial-state muon chirality and j the final-state tau chirality. We define θ as the angle between the incoming μ^- and outgoing τ^- momenta in the center-of-mass frame, so $\cos\theta > 0$ is the forward τ^- hemisphere. The vector and axial charge assignments are

$$\begin{aligned} \text{vector:} \quad & (g_L^\mu, g_R^\mu, g_L^\tau, g_R^\tau) = (g_X, g_X, -g_X, -g_X), \\ \text{axial:} \quad & (g_L^\mu, g_R^\mu, g_L^\tau, g_R^\tau) = (g_X, -g_X, -g_X, g_X). \end{aligned}$$

The sign reversal between the left- and right-handed couplings changes the helicity amplitudes and the relative weights of the $(1+\cos\theta)^2$ and $(1-\cos\theta)^2$ terms. Thus a total-rate reinterpretation of the vector result can obscure the axial-current effect, whereas the angular distribution retains sensitivity to the chiral structure.

For the same central angular acceptance $|\cos\theta| < 0.9$ used in the figure, the cut forward–backward asymmetry is

$$A_{\rm FB} = \frac{\displaystyle\int_0^{0.9} d\cos\theta\, \frac{d\sigma}{d\cos\theta} - \int_{-0.9}^{0} d\cos\theta\, \frac{d\sigma}{d\cos\theta}}{\displaystyle\int_{-0.9}^{0.9} d\cos\theta\, \frac{d\sigma}{d\cos\theta}}$$

. The asymmetry isolates the forward–backward odd part of the distribution and is therefore a chirality-sensitive observable that does not require polarized initial beams. The fully inclusive definition corresponds to replacing the cut value 0.9 by 1. In the benchmark shown in the paper, $\sqrt{s} = 3\,\text{TeV}$, $m_{A'} = 1\,\text{TeV}$, and $g_X = 0.05$, the vector benchmark changes the total rate relative to the SM prediction in $|\cos\theta| < 0.9$ by about 5%, whereas the axial benchmark changes it by about 2%. The finite-width prescription used for the plotted pure-axial curve is $\Gamma_{A'} = g_X^2 m_{A'}/(4\pi)$, which gives $\Gamma_{A'} \simeq 0.60\,\text{GeV}$ at the resonance $m_{A'} = \sqrt{s} = 3\,\text{TeV}$. In contrast, the axial benchmark produces $\Delta A_{\rm FB} \simeq -2.9 \times 10^{-2}$, larger in magnitude than the vector shift for the same benchmark. This comparison demonstrates why an angular observable is more discriminating than a simple total-rate rescaling for the axial interaction. Writing

$$S_+ = |\mathcal{A}_{LL}|^2 + |\mathcal{A}_{RR}|^2, \qquad S_- = |\mathcal{A}_{LR}|^2 + |\mathcal{A}_{RL}|^2,$$

the angular integration gives

$$A_{\rm FB}(c_{\rm max}) = \frac{c_{\rm max}}{1 + c_{\rm max}^2/3}\,\frac{S_+ - S_-}{S_+ + S_-}.$$

For the plotted cut $c_{\max} = 0.9$, the purely kinematic prefactor is approximately 0.709. Thus $A_{\rm FB}$ directly compares the same-chirality and opposite-chirality helicity combinations.

Polarization observables would provide complementary information because they project separately onto initial- and final-state chiral amplitudes. Following the convention used in Ref. [42], an initial-state left–right asymmetry can be defined with a polarized μ^- beam and an unpolarized μ^+ beam:

$$A_{\rm LR}^{\mu} = \frac{\sigma(\mu_L^- \mu^+ \to \tau^+\tau^-) - \sigma(\mu_R^- \mu^+ \to \tau^+\tau^-)}{\sigma(\mu_L^- \mu^+ \to \tau^+\tau^-) + \sigma(\mu_R^- \mu^+ \to \tau^+\tau^-)}.$$

With unpolarized initial beams, final-state tau polarimetry can instead measure

$$P_\tau = \frac{\sigma_{\tau_R} - \sigma_{\tau_L}}{\sigma_{\tau_R} + \sigma_{\tau_L}},$$

where

$$\sigma_{\tau_R} \equiv \sigma(\mu^+\mu^- \to \tau_R^-\tau_L^+), \qquad \sigma_{\tau_L} \equiv \sigma(\mu^+\mu^- \to \tau_L^-\tau_R^+)$$

in the relativistic limit. These observables resolve helicity combinations beyond the unpolarized $A_{\rm FB}$ distribution. We retain $A_{\rm FB}$ as the representative observable because the collider illustration in the manuscript uses unpolarized beams and does not model beam-polarization or tau-polarimetry efficiencies and systematic uncertainties. The polarization asymmetries are therefore identified as complementary observables, but no numerical reach is claimed.

The ee final state is not an equivalent tree-level channel in the pure-axial effective-current limit, because A' has no tree-level coupling to electrons. An electron coupling can arise through loop-induced kinetic mixing. The equal-mixing benchmark gives $|\epsilon_{\gamma A'}| \leq 2.12 \times 10^{-7}$ for $g_X \leq 0.1$ in the tabulated low-mass, on-shell range, corresponding to $e|\epsilon_{\gamma A'}| \lesssim 6.4 \times 10^{-8}$. This number is not a prediction for a multi-TeV muon collider: the mixing is momentum dependent, and the relevant virtuality for $\mu^+\mu^- \to e^+e^-$ is $q^2 = s$. A collider-level ee prediction would require evaluating $\epsilon_{\gamma A'}(s)$ with the corresponding matching condition. Thus, unlike the direct tree-level $\tau^+\tau^-$ channel, the ee channel is loop induced in the pure-axial limit.

We agree that S is an important complementary UV target for a future Higgs factory. The renormalizable portal operator $(H^\dagger H)(S^\dagger S)$ is allowed by the gauge symmetry; a nonzero renormalized portal coefficient induces Higgs–singlet mixing. For $m_s < \sqrt{s_{ee}} - m_Z$, this mixing permits associated production $e^+e^- \to Zs$, with a rate proportional to $\sin^2\theta_{hs}$. Separately, if $m_s < m_h/2$, the exotic decay $h \to ss$ can be kinematically open. Its rate is

controlled by the trilinear coupling g_{hss}, defined by $\mathcal{L} \supset -\frac{1}{2} g_{hss} h s^2$, and is not fixed by $\sin^2\theta_{hs}$ alone:

$$\Gamma(h \to ss) = \frac{g_{hss}^2}{32\pi m_h} \sqrt{1 - \frac{4m_s^2}{m_h^2}}.$$

Both g_{hss} and θ_{hs} depend on the scalar potential and portal parameter. For the same small-portal matching benchmark specified under Major Point 1.3, the negligible-mixing description applies away from the narrow near-degenerate region identified there; that region is not covered by the small-mixing approximation. We therefore identify $e^+e^- \to Zs$ and $h \to ss$ as conditional complementary channels, without assigning a model-specific rate or reach or asserting negligible mixing throughout the scalar-mass range. The scalar decay pattern, including $s \to A'A'$ when allowed, further determines the visible final states. The Higgs-factory scalar programme is therefore complementary to, rather than replaced by, the muon-collider angular test of the A' current.

Accordingly, the muon-collider result is presented as a concrete example of a future chirality-sensitive test, not as a detector-level discovery or exclusion projection. Its purpose is to show that the axial interaction has a distinctive angular signature that is not captured by a vector-to-axial total-rate reinterpretation, while the scalar S remains a separate target for a future Higgs factory.

Minor Point 6).

Terminology. The following points could be improved for clarity.

Minor Point 6.1).

The word "light" on page 2 has no reference scale.

Reply. The adjective "light" is not used as a quantitative mass classification. The Abstract states the broader MeV-to-TeV range, while the explicit comparison window in Fig. 2 is

$$10^{-3}\,\mathrm{GeV} \le m_{A'} \le 10^{2}\,\mathrm{GeV}, \qquad 10^{-6} \le g_X \le 1.$$

These numerical ranges, together with the observables included in the analysis, define the phenomenological domain; the word "light" does not introduce an additional universal mass threshold.

Minor Point 6.2).

> *At the bottom of page 11, when referring to Table II, the authors write "representative bounds." What does representative mean? Does it mean these are the most stringent bounds? Are these "some" bounds? Is the list "complete" in any sense?*

Reply. Here, "representative" means selected, non-exhaustive, and source-dependent. The entries are selected to illustrate distinct scalar-mass regimes and physical mechanisms; they are neither a complete inventory of the literature nor necessarily the strongest constraint at every mass. Their numerical values and mass intervals retain the coupling normalization and source assumptions of the corresponding analyses, including the relevant production and free-streaming/trapping conditions. Table II also distinguishes direct scalar–muon Yukawa limits from loop-induced fifth-force and photon-mediated results, whose translation requires the matching described under Major Point 1 [1, 3, 7, 4, 5, 6]. The entries are not combined into a global exclusion of the UV parameter space.

Minor Point 6.3).

> *Between page 12–13: "...do not provide a universal g_X bound, but they show that the intermediate scalar-mass region cannot simply be assumed unconstrained." What does intermediate mass mean? It is also not clear what the intended content of this statement is: are the authors presenting a bound? Why would the reader assume that a new particle in this range would be unconstrained?*

Reply. The intermediate-mass range refers specifically to the direct BaBar and Belle searches for

$$e^+e^- \to \tau^+\tau^- s, \qquad s \to e^+e^- \text{ or } \mu^+\mu^-,$$

which cover approximately $0.04\,\text{GeV} \lesssim m_s \lesssim 6.5\text{–}7\,\text{GeV}$ [8, 9]. These searches provide direct experimental coverage of the intermediate-mass scalar region. For the present minimal UV benchmark with negligible portal mixing, the tree-level radial-scalar Yukawa couplings are $y_{s\ell\ell} = m_\ell/v_s$ for $\ell = \mu, \tau$. Accordingly, the UV translation uses the dimuon channel above threshold; the generic low-mass $s \to e^+e^-$ search channel is not treated as a direct constraint on this

benchmark. In the mass-proportional benchmark, the Belle limit $\xi^{\rm lim} = 0.483$ at $m_s = 1\,\text{GeV}$ [9] gives

$$v_s > \frac{v_h}{0.483} \simeq 509\,\text{GeV}, \qquad y_{s\mu\mu} < 2.1 \times 10^{-4},$$

and, using $m_{A'} = 2g_X v_s$,

$$g_X < 9.8 \times 10^{-4} \left(\frac{m_{A'}}{\text{GeV}}\right)$$

at 90% confidence level for the illustrative $m_s = 1\,\text{GeV}$ point with $m_{A'} > 0.5\,\text{GeV}$, where $s \to A'A'$ is closed. This is separate from the $m_s \gtrsim 10\,\text{GeV}$ reference benchmark used for Fig. 2 and is not a universal bound on g_X; for $m_{A'} < 0.5\,\text{GeV}$, the visible branching fraction must be treated as described under Major Point 1.

The Belle II references cover two distinct analyses and should not be described collectively as spin-one-only searches. The $e^+e^- \to \mu^+\mu^- X$, $X \to \tau^+\tau^-$ analysis covers 3.6–10 GeV and explicitly includes a mass-proportional spin-zero leptophilic scalar [10]; the four-muon analysis covers 0.212–9 GeV and also tests a muonphilic scalar [11]. The former has the same charged-lepton Yukawa scaling as the present radial scalar, while the latter is a complementary muonphilic benchmark. For the UV completion considered here, the additional $s \to A'A'$ width modifies the visible dilepton or ditau branching fractions when the channel is open. The Belle II scalar results therefore provide relevant published scalar sensitivity, but their coupling and $\sigma \times \text{BR}$ limits translate into the present UV parameters only after using the model-specific total width and visible branching fractions, as detailed under Major Point 1.

Minor Point 6.4).

> *Below (34): "we impose Eq. (34) point by point", it's not clear what is meant by "point."*

Reply. A "point" denotes one grid point $(m_{A'}, g_X)$ in the parameter scan shown in Fig. 2. At each point, the symmetry-breaking scale is derived from

$$v_s = \frac{m_{A'}}{2g_X},$$

rather than varied independently. For a specified scalar mass m_s, the scalar self-coupling is then determined by

$$\lambda_S = \frac{m_s^2}{2v_s^2} = \frac{2g_X^2 m_s^2}{m_{A'}^2}.$$

The condition

$$\lambda_S \leq \lambda_S^{\max}$$

is consequently checked at each scan point for the adopted scalar benchmark.

Equation (34) is checked point by point to test the theoretical consistency of the chosen scalar benchmark; it is not itself an experimental exclusion. The experimental constraints are a separate step.

Minor Point 6.5).

> *Page 21, first paragraph of Section E. What is the "parameter region of interest"? This wasn't clearly defined anywhere.*

Reply. We agree that the phrase "parameter region of interest" did not specify the mass coverage of the searches considered in Section III.E. The relevant four-muon searches cover approximately

$$\begin{aligned} \text{BaBar}: &\quad 0.212\text{–}10\,\text{GeV}, \\ \text{Belle II}: &\quad 0.212\text{–}9\,\text{GeV}, \\ \text{CMS}: &\quad 5\text{–}70\,\text{GeV}. \end{aligned}$$

These intervals describe the mass coverage of the published analyses; they do not imply that the entire intervals are excluded in our model.

The relevant processes are

$$e^+e^- \to \mu^+\mu^- A', \qquad A' \to \mu^+\mu^-,$$

and

$$Z \to \mu^+\mu^- A', \qquad A' \to \mu^+\mu^-.$$

The reinterpretation accounts for the production rate and $\mathrm{Br}(A' \to \mu^+\mu^-)$. The experimental analyses also impose prompt-resonance selection requirements. Their limits are applicable to our model only where the predicted A' decay length is compatible with those requirements; the mass coverage alone does not establish this condition throughout the parameter space.

As an illustrative check, at $m_{A'} = 0.212\,\text{GeV}$ and $g_X = 7 \times 10^{-4}$, the neutrino decay channels alone give $c\tau_{A'} \lesssim 0.072\,\mu\text{m}$, corresponding to a laboratory-frame decay length of approximately $1.8\,\mu\text{m}$ for $\gamma \simeq 25$. This estimate supports the prompt-decay approximation at this benchmark point, but does not establish its validity throughout the full parameter space.

The vector-to-axial rate-matching approximation is controlled for $m_{A'} \gg m_\mu$. Near the dimuon threshold, a rate-level comparison alone is insufficient to infer a detector-level exclusion.

Minor Point 6.6).

> *Page 21, end of section D: What's the significance of the* $v_s = 2\,\mathrm{GeV}$ *line? What does it mean that* $y_{s\mu\mu}$ *"becomes large" (non-perturbative)? Why plot it? Is it a perturbativity bound?*

Reply. The line at $v_s \simeq 2\,\mathrm{GeV}$ follows from the scalar self-coupling condition for the reference benchmark $m_s = 10\,\mathrm{GeV}$ and $\lambda_S^{\max} = 4\pi$:

$$v_s \geq \frac{m_s}{\sqrt{2\lambda_S^{\max}}} = \frac{10\,\mathrm{GeV}}{\sqrt{8\pi}} \simeq 1.995\,\mathrm{GeV}.$$

At this reference value, $y_{s\mu\mu} = m_\mu/v_s \simeq 0.053$, so $y_{s\mu\mu}^2/(4\pi) \simeq 2.2 \times 10^{-4} \ll 1$. The line is therefore a scalar-quartic consistency boundary for the stated m_s benchmark, not a Yukawa nonperturbativity limit or an experimental exclusion. Its relation to the point-by-point scalar-mass condition is given in Major Point 1.2 and Major Point 2.

Summary:

The paper constructs an anomaly-free axial $U_A(1)_{L_{\mu-\tau}}$ UV completion in which the scalar S and vector-like leptons generate the charged-lepton masses. The present response makes explicit how the scalar mass, its lepton couplings, perturbativity, direct scalar searches, and its contribution to $(g-2)_\mu$ fit together, while identifying the benchmark under which Fig. 2 isolates the axial-current phenomenology.

In the dedicated response to Major Point 3, we discuss an established two-singlet type-I seesaw as a possible gauge-invariant completion of the neutrino sector: the existing charge-two scalar S, an additional charge-one scalar σ, and three right-handed neutrinos can generate a general Majorana mass matrix and realistic mixing. This is a response to the referee's neutrino-oscillation concern, not a change to the field content of the original paper. The original S + VLL one-VEV model remains the baseline for the scalar, axial-current, kinetic-mixing, finite-vector, and stellar-cooling discussions above. The mass-texture argument is not a global UV parameter-space analysis.

For the additional constraints, the response gives a momentum-dependent kinetic-mixing estimate and compares the leading finite-vector effects in trident production and $(g-2)_\mu$. The Table II entries are source-dependent scalar–muon benchmarks rather than a full-UV exclusion, whereas the A' supernova discussion concerns a distinct vector-current problem: neutrino-coalescence production and invisible widths follow from the unchanged $L_\mu - L_\tau$ neutrino couplings, while semi-Compton production and absorption depend on the axial charged-muon current; the cited vector-model curve remains a reference scale, and Fig. 1 adds a separately evaluated, explicitly conditional axial-EFT ballistic-cooling exclusion. All conclusions remain tied to the explicitly stated benchmarks and do not constitute a global exclusion of the full UV parameter space.

Sincerely yours,

Rundong Fang, Jinhui Guo, Ming Li, Jia Liu, Xiao-Ping Wang, Yiheng Xiong

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